United States Patent Application Publication	20250279153
Kind Code	A2
Publication Date	September 04, 2025
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SPARING TECHNIQUES IN STACKED MEMORY ARCHITECTURES

Abstract

Methods, systems, and devices for sparing techniques in stacked memory architectures are described. A memory system may implement a stacked memory architecture that includes a set of array dies stacked along a direction and a logic die coupled with the set of array dies. Each array die may include one or more memory arrays accessible using one or more first interface blocks of the array die. To support sparing, the memory system may remap access from one or more first memory arrays of the set of array dies to one or more second memory arrays of the set of array dies. Logic circuitry of the logic die may be operable to perform the remapping in accordance with one or more levels of granularity, such as at a die level, channel level, pseudo-channel level, bank level, or a combination thereof.

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Family ID: **92141860**

Appl. No.: **18/775981**

Filed: **July 17, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20250149108 A1	May. 08, 2025

Related U.S. Application Data

us-provisional-application US 63595650 20231102

Publication Classification

Int. Cl.: **G11C29/00** (20060101); **G11C29/52** (20060101)

U.S. Cl.:

CPC **G11C29/76** (20130101); **G11C29/52** (20130101); **G11C29/883** (20130101);

Background/Summary

CROSS REFERENCE [0001] The present Application for Patent claims the benefit of U.S. Provisional Patent Application No. 63/595,650 by Mylavarapu et al., entitled “SPARING TECHNIQUES IN STACKED MEMORY ARCHITECTURES,” filed Nov. 2, 2023, which is assigned to the assignee hereof, and which is expressly incorporated by reference herein.

TECHNICAL FIELD
[0002] The following relates to one or more systems for memory, including sparing techniques in stacked memory architectures.
BACKGROUND
[0003] Memory devices are used to store information in devices such as computers, user devices, wireless communication devices, cameras, digital displays, and others. Information is stored by programming memory cells within a memory device to various states. For example, binary memory cells may be programmed to one of two supported states, often denoted by a logic 1 or a logic 0. In some examples, a single memory cell may support more than two states, any one of which may be stored by the memory cell. To store information, a memory device may write (e.g., program, set, assign) states to the memory cells. To access stored information, a memory device may read (e.g., sense, detect, retrieve, determine) states from the memory cells.
[0004] Various types of memory devices exist, including magnetic hard disks, random access memory (RAM), read-only memory (ROM), dynamic RAM (DRAM), synchronous dynamic RAM (SDRAM), static RAM (SRAM), ferroelectric RAM (FeRAM), magnetic RAM (MRAM), resistive RAM (RRAM), flash memory, phase change memory (PCM), self-selecting memory, chalcogenide memory technologies, not-or (NOR) and not-and (NAND) memory devices, and others. Memory cells may be described in terms of volatile configurations or non-volatile configurations. Memory cells in a non-volatile configuration may maintain stored logic states for extended periods of time even in the absence of an external power source. Memory cells in a volatile configuration may lose stored states when disconnected from an external power source.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] FIG. **1** shows an example of a system that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein.
[0006] FIG. **2** shows an example of a system that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein.
[0007] FIG. **3** shows an example of an interface architecture that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein.
[0008] FIG. **4** shows an example of a manufacturing process that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein.

[0009] FIGS. **5**A, **5**B, **5**C, and **5**D show examples of sparing diagrams that support sparing techniques in stacked memory architectures in accordance with examples as disclosed herein.
[0010] FIG. **6** shows a block diagram of a system that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein.
[0011] FIG. **7** shows a block diagram of a logic die that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein.
[0012] FIGS. **8** and **9** show flowcharts illustrating methods that support sparing techniques in stacked memory architectures in accordance with examples as disclosed herein.

DETAILED DESCRIPTION

[0013] Some memory systems may include a stack of semiconductor dies, including one or more memory dies (which may be referred to as array dies) above a logic die that is operable to access a set of memory arrays distributed across the one or more memory dies. Such a stacked architecture may be implemented as part of a coupled dynamic random access memory (DRAM) system, and may support solutions for memory-centric logic, such as graphics processing units (GPUs), among other implementations. In some examples, a 3D stacked memory system may be closely coupled (e.g., physically coupled, electrically coupled) with a processor, such as a GPU or other host, as part of a physical memory map accessible to the processor. Such coupling may include one or more processors being implemented in a same semiconductor die as at least a portion of a 3D stacked memory system (e.g., as part of a logic die), or a processor being implemented in a die that is directly coupled (e.g., fused) with another die that includes at least a portion of a 3D stacked memory system. Unlike cache-based memory, 3D stacked memory may not be backed by a level of external memory with the same physical addresses. For example, a 3D stacked memory may be associated with and located within a dedicated base address, where each portion of the 3D stacked memory may be non-overlapping within the address.
[0014] To manufacture a memory system that implements a stacked architecture (e.g., a 3D stacked memory system), the granularity at which semiconductor dies are stacked (e.g., and coupled) may be at the wafer level. For example, due to the close coupling of the stacked semiconductor dies, semiconductor wafers that include multiple semiconductor dies may be stacked and coupled (e.g., bonded) together, and respective stacks of coupled dies may be individually separated (e.g., cut, divided) from the stack of semiconductor wafers (e.g., rather than selecting individual semiconductor dies for stacking and coupling). Wafers with a relatively high yield of dies, such as wafers that include relatively fewer dies with defects (e.g., failures), may be selected for inclusion in a wafer stack. However, a non-perfect yield of the wafers may be compounded in the stack of wafers, such that a yield of a resulting die stack separated from the wafer stack may be reduced. For example, an eight-high array die stack that is separated from a stack of wafers that each have approximately 90% yield may have a yield of approximately 40%. As such, performance (e.g., latency, bandwidth, storage capacity) or yield of memory systems may be adversely affected by such stacking.
[0015] In accordance with techniques described herein, a memory system that implements a stacked architecture may support sparing techniques to improve yield of an array die stack (e.g., compensate for relatively lower yield of the array die stack). For example, one or more spare arrays or dies may be included in the array die stack, and a logic die may include logic circuitry to support sparing (e.g., replacing) one or more failed components of the array dies (e.g., a bank, a pseudo-channel, a channel, an array, an array die) with one or more components of the one or more spare arrays or dies (e.g., a spare bank, a spare-pseudo-channel, a spare channel, a spare array, a spare array die). For instance, the logic die may include interface blocks (e.g., memory interface blocks (MIBs)) that are used to access one or more memory arrays of the array dies. The logic circuitry may remap accessing of the failed components to the components of the one or more spare dies such that accesses to an address space corresponding to the failed components (e.g., using one or more first interface blocks) may instead access the one or more components of the one or more

spare dies (e.g., using one or more second interface blocks) in accordance with the remapping. Additionally, or alternatively, one or more array dies of the array die stack may include spare portions to which the logic circuitry may remap accessing of failed components.

[0016] Supporting sparing in accordance with the described techniques may increase the yield of a memory system that implements a stacked architecture, such as a 3D stacked memory system, for example, by supporting a replacement of failed components with spare components. As such, performance of the memory system may be improved. Additionally, a time-to-market of products that implement 3D stacked memory systems may be reduced. For example, products released to the market may be associated with yield constraints, and improving yield via sparing techniques described herein may enable such constraints to be satisfied without improving manufacturing techniques to increase the yield of individual semiconductor wafers, which may involve a delay before such improvements are realized. Additionally, or alternatively, sparing techniques described herein may reduce costs as a final yield of an array die stack may be unknown until after it is coupled (e.g., wire-to-wire bonded) with a host system (e.g., included on a logic die). Accordingly, if the yield does not meet yield constraints, the entire system, including the logic die and array dies, may be discarded. Thus, improving array die stack yield using sparing techniques described herein may increase the likelihood that yield constraints are met, thereby reducing costs of manufacturing 3D stacked memory systems.

[0017] Features of the disclosure are illustrated and described in the context of systems and dies. Features of the disclosure are further illustrated and described in the context of an interface architecture, a manufacturing process, sparing diagrams, block diagrams, and flowcharts.

[0018] FIG. **1** illustrates an example of a system **100** that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein. The system **100** may include portions of an electronic device, such as a computing device, a mobile computing device, a wireless communications device, a graphics processing device, a vehicle, or other systems. The system **100** includes a host system **105**, a memory system **110**, and one or more channels **115** coupling the host system **105** with the memory system **110** (e.g., to provide a communicative coupling). The system **100** may include one or more memory systems **110**, but aspects of the one or more memory systems **110** may be described in the context of a single memory system **110**.

[0019] The host system **105** may be an example of a processor (e.g., circuitry, processing circuitry, one or more processing components) that uses memory to execute processes, such as a processing system of a computing device, a mobile computing device, a wireless communications device, a graphics processing device, a wearable device, an internet-connected device, a vehicle controller, a system on a chip (SoC), or other stationary or portable electronic device, among other examples. The host system **105** may include one or more of an external memory controller **120**, a processor **125**, a basic input/output system (BIOS) component **130**, or other components (e.g., a peripheral component, an input/output controller, not shown). The components of the host system **105** may be coupled with one another using a bus **135**.

[0020] An external memory controller **120** may be configured to enable communication of information (e.g., data, commands, control information, configuration information) between components of the system **100** (e.g., between components of the host system **105**, such as the processor **125**, and the memory system **110**). An external memory controller **120** may process (e.g., convert, translate) communications exchanged between the host system **105** and the memory system **110**. In some examples, an external memory controller **120**, or other component of the system **100**, or associated functions described herein, may be implemented by or be part of the processor **125**. For example, an external memory controller **120** may be hardware, firmware, or software (e.g., instructions), or some combination thereof implemented by a processor **125** or other component of the system **100** or the host system **105**. Although an external memory controller **120** is illustrated outside the memory system **110**, in some examples, an external memory controller **120**, or its functions described herein, may be implemented by one or more components of a memory system **110** (e.g., a memory system controller **155**, a local memory controller **165**) or vice versa. In various examples, the host system **105** or an external memory controller **120** may be referred to as a host.

[0021] A processor **125** may be operable to provide functionality (e.g., control functionality) for the system **100** or the host system **105**. A processor **125** may be a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof. In some examples, a processor **125** may be an example of a central processing unit (CPU), a graphics processing unit (GPU), a general-purpose GPU (GPGPU), or an SoC, among other examples.

[0022] In some examples, the system **100** or the host system **105** may include an input component, an output component, or a combination thereof. Input components may include a sensor, a microphone, a keyboard, another processor (e.g., on a printed circuit board), an interface (e.g., a user interface, an interface between other devices), or a peripheral that interfaces with system **100** via one or more peripheral components, among other examples. Output components may include a display, audio speakers, a printing device, another processor on a printed circuit board, or a peripheral that interfaces with the system **100** via one or more peripheral components, among other examples.

[0023] The memory system **110** may be a component of the system **100** that is operable to provide physical memory locations (e.g., addresses) that may be used or referenced by the system **100**. The memory system **110** may include a memory system controller **155** and one or more memory dies **160** (e.g., memory chips) to support a capacity for data storage. The memory system **110** may be configurable to work with one or more different types of host systems **105**, and may respond to and execute commands provided by the host system **105** (e.g., via an external memory controller **120**). For example, the memory system **110** (e.g., a memory system controller **155**) may receive a write command indicating that the memory system **110** is to store data received from the host system **105**, or receive a read command indicating that the memory system **110** is to provide data stored in a memory die **160** to the host system **105**, or receive a refresh command indicating that the memory system **110** is to refresh data stored in a memory die **160**, among other types of commands and operations.

[0024] A memory system controller **155** may include components (e.g., circuitry, logic) operable to control operations of the memory system **110**. A memory system controller **155** may include hardware, firmware, or instructions that enable the memory system **110** to perform various operations, and may be operable to receive, transmit, or execute commands, data, or control information related to operations of the memory system **110**. A memory system controller **155** may be operable to communicate with one or more of an external memory controller **120**, one or more memory dies **160**, or a processor **125**. In some examples, a memory system controller **155** may control operations of the memory system **110** in cooperation with a local memory controller **165** of a memory die **160**.

[0025] Each memory die **160** may include a local memory controller **165** and a memory array **170**. A memory array **170** may be a collection of memory cells, with each memory cell being operable to store one or more bits of data. A memory die **160** may include a two-dimensional (2D) array of memory cells, or a three-dimensional (3D) array of memory cells. In some examples, a 2D memory die **160** may include a single memory array **170**. In some examples, a 3D memory die **160** may include two or more memory arrays **170**, which may be stacked or positioned beside one another (e.g., relative to a substrate).

[0026] A local memory controller **165** may include components (e.g., circuitry, logic) operable to control operations of a memory die **160**. In some examples, a local memory controller **165** may be operable to communicate (e.g., receive or transmit data or commands or both) with a memory system controller **155**. In some examples, a memory system **110** may not include a memory system controller **155**, and a local memory controller **165** or an external memory controller **120** may perform various functions described herein. As such, a local memory controller **165** may be operable to communicate with a memory system controller **155**, with other local memory controllers **165**, or directly with an external memory controller **120**, or a processor **125**, or any combination thereof. Examples of components that may be included in a memory system controller **155** or a local memory controller **165** or both may include receivers for receiving signals (e.g., from the external memory controller **120**), transmitters for transmitting signals (e.g., to the external memory controller **120**), decoders for decoding or demodulating received signals, encoders for encoding or modulating signals to be transmitted, sense components for sensing states of memory cells of a memory array **170**, write components for writing states to memory cells of a memory array **170**, or various other components operable for supporting described operations of a memory system **110**.

[0027] A host system **105** (e.g., an external memory controller **120**) and a memory system **110** (e.g., a memory system controller **155**) may communicate information (e.g., data, commands, control information, configuration information) using one or more channels **115**. Each channel **115** may be an example of a transmission medium that carries information, and each channel **115** may include one or more signal paths (e.g., a transmission medium, an electrical conductor, an electrically conductive path) between terminals associated with the components of the system **100**. For example, a channel **115** may be associated with a first terminal (e.g., including one or more pins, including one or more pads) at the host system **105** and a second terminal at the memory system **110**. A terminal may be an example of a conductive input or output point of a device of the system **100**, and a terminal may be operable to act as part of a channel **115**. In some implementations, at least the channels **115** between a host system **105** and a memory system **110** may include or be referred to as a host interface (e.g., a physical host interface). In some implementations, a host interface may include or be associated with interface circuitry (e.g., signal drivers, signal latches) at the host system **105** (e.g., at an external memory controller **120**), or at the memory system **110** (e.g., at a memory system controller **155**), or both.

[0028] In some examples, a channel **115** (e.g., associated signal paths and terminals) may be dedicated to communicating one or more types of information. For example, the channels **115** may include one or more command and address channels, one or more clock signal channels, one or more data channels, among other channels or combinations thereof. In some examples, signaling may be communicated via the channels **115** using single data rate (SDR) signaling or double data rate (DDR) signaling. In SDR signaling, one modulation symbol (e.g., signal level) of a signal may be registered for each clock cycle (e.g., on a rising or falling edge of a clock signal). In DDR signaling, two modulation symbols of a signal may be registered for each clock cycle (e.g., on both a rising edge and a falling edge of a clock signal).

[0029] In some examples, at least a portion of the system **100** may implement a stacked die architecture in which multiple semiconductor dies are physically and communicatively coupled. In some implementations, one or more semiconductor dies may include multiple instances of interface circuitry (e.g., of a memory system **110**, memory interface blocks) that are each associated with accessing a respective set of one or more memory arrays **170** of one or more other semiconductor dies. In some cases, circuitry for accessing one or more memory arrays **170** may be distributed among multiple semiconductor dies of a stack (e.g., a stack of multiple directly-coupled semiconductor dies). For example, a first die may include a logic block (e.g., a common logic block, a central logic block, logic circuitry) operable to configure a set of multiple first interface blocks (e.g., MIBs, instances of first interface circuitry) of the first die. In some examples, the system may include a respective controller (e.g., a memory controller, a host interface controller, at least a portion of a memory system controller **155**, at least a portion of an external memory controller **120**, or a combination thereof) for each first interface block to support access operations (e.g., to access one or more memory arrays **170**) via the first interface block. The system **100** may also include non-volatile storage, one or more sensors, or a combination thereof to support various operations of the system **100**.

[0030] In some examples, multiple semiconductor dies of a memory system **110** (e.g., a 3D stacked memory system) may include one or more array dies (e.g., memory dies **160**) stacked with a logic die (e.g., that includes the host system **105**, that is coupled with another die that includes the host system **105**) that includes interface blocks operable to access a set of memory arrays **170** distributed across the one or more second dies. In some cases, one or more components of an array die may fail (e.g., be defective, malfunction) such that successful accessing of one or more of the memory arrays **170** of the array die using an interface block of the logic die may be unreliable or impossible. As such, a yield of the array die may be reduced. Yield reductions may be compounded in a memory system **110** that includes a stack of array dies. For example, to manufacture such a memory system **110**, semiconductor wafers may be stacked and coupled (e.g., bonded) together, and respective stacks of array dies may be individually separated (e.g., cut out) from the stack of semiconductor wafers (e.g., rather than selecting individual semiconductor dies for stacking and coupling). Although wafers with a relatively high yield may be selected for inclusion in such a wafer stack, a non-perfect yield of the wafers may be compounded in the stack of wafers such that a yield of a resulting die stack separated from the wafer stack may be reduced. As a result, performance (e.g., latency, bandwidth, storage capacity) or yield of the memory system **110** may be adversely affected by such stacking.

[0031] In accordance with techniques described herein, a system **100** that implements a stacked architecture may support sparing techniques to improve yield of an array die stack (e.g., a memory system **110**). For example, one or more spare array dies may be included in the array die stack, and a logic die may include logic circuitry to support sparing (e.g., replacing) one or more failed components of the array dies (e.g., a bank, a pseudo-channel, a channel, an array, an array die) with one or more components of the one or more spare dies (e.g., a spare bank, a spare-pseudo-channel, a spare channel, a spare array, a spare array die). For instance, the logic circuitry may remap accessing of the failed components to the components of the one or more spare dies such that accesses to an address space (e.g., available to a host system **105**) corresponding to the failed components may instead access the one or more components of the one or more spare dies in accordance with the remapping. Additionally, or alternatively, one or more array dies of the array die stack may include spare portions to which the logic circuitry may remap accessing of failed components.

[0032] In addition to applicability in systems as described herein, techniques for sparing in stacked memory architectures may be generally implemented to support artificial intelligence applications. As the use of artificial intelligence increases to support machine learning, analytics, decision making, or other related applications, electronic devices that support artificial intelligence applications and processes may be desired. For example, artificial intelligence applications may be associated with accessing relatively large quantities of data for analytical purposes and may benefit from memory devices capable of effectively and efficiently storing relatively large quantities of data or accessing stored data relatively quickly. Implementing the techniques described herein may support artificial intelligence and/or machine learning techniques by improving memory access bandwidth and increasing memory capacity, for example, by supporting the replacement of failed components with spare components to maintain bandwidth and capacity or mitigate bandwidth and capacity loss despite the failure of the components, among other benefits.

[0033] FIG. **2** illustrates an example of a system **200** (e.g., a semiconductor system, a system of coupled semiconductor dies) that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein. The system **200** illustrates an example of a die **205** (e.g., a semiconductor die, a host die, a processor die, a logic die) that is coupled with one or more dies **240** (e.g., dies **240-*a*-1** and **240-*a*-2**, semiconductor dies, memory dies, array dies). A die **205** or a die **240** may be formed using a respective semiconductor substrate (e.g., a substrate of crystalline semiconductor material such as silicon, germanium, silicon-germanium, gallium arsenide, or gallium nitride), or a silicon-on-insulator (SOI) substrate (e.g., silicon-on-glass (SOG), silicon-on-sapphire (SOS)), or epitaxial semiconductor materials formed on another substrate, among other examples. Although the illustrated example of a system **200** includes two dies **240**, a system **200** in accordance with the described techniques may include any quantity of one or more dies **240** coupled with a die **205**. Further, although non-limiting examples of the system **200** herein are generally described in terms of applicability to memory systems, memory sub-systems, memory devices, or a combination thereof, examples of the system **200** are not so limited. For example, aspects of the present disclosure may be applied as well to any computing system, computing sub-system, processing system, processing sub-system, component, device, structure, or other types of systems or sub-systems used for applications such as data collecting, data processing, data storage, networking, communication, power, artificial intelligence, system-on-a-chip, control, telemetry, sensing and monitoring, digital entertainment, or any combination thereof.

[0034] The system **200** illustrates an example of interface circuitry between a host and memory (e.g., via a host interface, via a physical host interface) that is implemented in (e.g., divided between) multiple semiconductor dies (e.g., a stack of directly coupled dies). For example, the die **205** may include a set of one or more interface blocks **220** (e.g., interface blocks **220-*a*-1** and **220-*a*-2**, memory interface blocks), and each die **240** may include a set of one or more interface blocks **245** and one or more memory arrays **250** (e.g., die **240-*a*-1** including an interface block **245-*a*-1** coupled with a set of one or more memory arrays **250-*a*-1**, die **240-*a*-2** including an interface block **245-*a*-2** coupled with a set of one or more memory arrays **250-*a*-2**). The memory arrays **250** may be examples of memory arrays **170**, and may include memory cells of various architectures, such as RAM, DRAM, SDRAM, SRAM, FeRAM, MRAM, RRAM, PCM, chalcogenide, NOR, or NAND memory cells, or any combination thereof.

[0035] Although the example of system **200** is illustrated with one interface block **245** included in each die **240**, a die **240** in accordance with the described techniques may include any quantity of one or more interface blocks **245**, each coupled with a respective set of one or more memory arrays **250**, and each coupled with a respective interface block **220** of a die **205**. Thus, the interface circuitry of a system **200** may include one or more interface blocks **220** of a die **205**, with each interface block **220** being coupled with (e.g., in communication with) a corresponding interface block **245** of a die **240** (e.g., external to the die **205**). In some examples, a coupled combination of an interface block **220** and an interface block **245** (e.g., coupled via a bus associated with one or more channels, such as one or more data channels, one or more control channels, one or more clock channels, or a combination thereof) may include or be referred to as a data path associated with a respective set of one or more memory arrays **250**.

[0036] In some implementations, the die **205** may include a host processor **210**. A host processor **210** may be an example of a host system **105**, or a portion thereof (e.g., a processor **125**, aspects of an external memory controller **120**, or both). The host processor **210** may be configured to perform operations that implement storage of the memory arrays **250**. For example, the host processor **210** may receive data read from the memory arrays **250**, or may transmit data to be written to the memory arrays **250**, or both (e.g., in accordance with an application or other operations of the host processor **210**). Additionally, or alternatively, a host processor **210** may be external to a die **205**, such as in another semiconductor die or other component that is coupled with (e.g., communicatively coupled with, directly coupled with, bonded with) the die **205** via one or more contacts **212**.

[0037] A host processor **210** may be configured to communicate (e.g., transmit, receive) signaling with the interface blocks **220** via one or more host interfaces **216** (e.g., a physical host interface), which may implement aspects of channels **115** described with reference to FIG. **1**. For example, the host processor **210** may be configured to transmit access signaling (e.g., control signaling, access command signaling, configuration signaling) via host interfaces **216**, which may be received by the interface blocks **220** to support access operations (e.g., read operations, write operations) on the memory arrays **250**. In some examples, a host interface **216** may include a respective set of one or more signal paths for each interface block **220**, such that the host processor **210** may communicate with each interface block **220** via the respective set of signal paths (e.g., in accordance with a selection of the respective set to perform access operations via an interface block **220** that is selected by the host processor **210**). Additionally, or alternatively, a host interface **216** may include one or more signal paths that are shared among multiple interface blocks **220** (not shown), and an interface block **220**, or a host processor **210**, or both may interpret, ignore, respond to, or inhibit response to signaling via shared signal paths of the host interface **216** based on a logical indication (e.g., an addressing indication associated with the interface block **220** or an interface enable signal, which may be provided by the host processor **210** or the corresponding interface block **220**, depending on signaling direction).

[0038] In some examples, a respective host interface **216** may be coupled between a set of one or more interface blocks **220** and a controller **215** (e.g., host interface **216-*a*-1** coupled between interface block **220-*a*-1** and controller **215-*a*-1**, host interface **216-*a*-2** coupled between interface block **220-*a*-2** and controller **215-*a*-2**). The one or more interface blocks **220** and the controller **215** may communicate (e.g., collaborate) to perform one or more operations (e.g., scheduling operations, access operations, operations initiated by a host processor **210**) associated with a memory array **250**. A controller **215** may be an example of control circuitry (e.g., memory controller circuitry, host interface control circuitry), and may be associated with implementing respective instances of one or more aspects of an external memory controller **120**, or of a memory system controller **155**, or a combination thereof for each interface block **220**. In some examples, controllers **215** may be implemented in a die **205** whether a host processor **210** is included in the die **205**, or is external to the die **205**, and the interface block **220** may communicate with the host processor **210** via one or more controllers **215**. In some other examples, controllers **215** may be implemented external to a die **205** (e.g., in another die, not shown, coupled with respective interface blocks **220** via respective terminals for each of the respective host interfaces **216**), which may be in a same die as or a different die from a die that includes a host processor **210**. In some other examples, aspects of one or more controllers **215** may be included in the host processor **210**. Although the example of system **200** is illustrated as including a controller **215** for each interface block **220**, in various examples, a controller **215** may be coupled with any quantity of one or more interface blocks **220**, and a given interface block **220** may be operable based on single controller **215**, or by one or more of a set of multiple controllers **215** (e.g., in accordance with a controller multiplexing scheme).

[0039] In some examples, a host processor **210** may determine to access an address (e.g., a logical address of a memory array **250**, a physical address of a memory array **250**, an address of an interface block **220**), and determine which controller **215** to transmit access signaling to for accessing the address (e.g., a controller **215** or interface block **220** corresponding to the address). In some examples, the address may be associated with a row of memory cells of the memory array **250**. The host processor **210** may transmit access signaling to the determined controller **215** and, in turn, the determined controller **215** may transmit access signaling to the corresponding interface block **220**. The corresponding interface block **220** may subsequently transmit access signaling to the coupled interface block **245** to access the determined address (e.g., in a corresponding memory array **250**).

[0040] A die **205** may also include a logic block **230** (e.g., a shared logic block, a central logic block, common logic circuitry), which may be configured to communicate (e.g., transmit, receive) signaling with the interface blocks **220** of the die **205**. In some cases, the logic block **230** may be configured to communicate information (e.g., commands, instructions, indications, data) with one or more interface blocks **220** to facilitate operations of the system **200**. For example, a logic block **230** may be configured to transmit configuration signaling (e.g., initialization signaling, evaluation signaling, mapping signaling), which may be received by interface blocks **220** to support configuration of the interface blocks **220** or other aspects of operating the dies **240** (e.g., via the respective interface blocks **245**). A logic block **230** may be coupled with each interface block **220** via a respective bus **231** (e.g., bus **231-*a*-1** associated with the interface block **220-*a*-1**, bus **231-*a*-2** associated with the interface block **220-*a*-2**). In some examples, respective buses **231** may each include a respective set of one or more signal paths, such that a logic block **230** may communicate with each interface block **220** via the respective set of signal paths. Additionally, or alternatively, respective buses **231** may include one or more signal paths that are shared among multiple interface blocks **220** (not shown).

[0041] In some implementations, a logic block **230** may be configured to communicate (e.g., transmit, receive) signaling with a host processor **210** (e.g., via a bus **232**, via a contact **212** for a host processor **210** external to a die **205**) such that the logic block **230** may support an interface between the interface blocks **220** and the host processor **210**. For example, a host processor **210** may be configured to transmit initialization signaling (e.g., boot commands), or other configuration or operational signaling, which may be received by a logic block **230** to support initialization, configuration, or other operations of the interface blocks **220**. Additionally, or alternatively, in some implementations, a logic block **230** may be configured to communicate (e.g., transmit, receive) signaling with a component outside the system **200** via a bus **233** (e.g., and via a contact **234**), such that the logic block **230** may support an interface that bypasses a host processor **210**. Additionally, or alternatively, a logic block **230** may communicate with a host processor **210**, and may communicate with one or more memory arrays **250** of one or more dies **240** (e.g., to perform self-test operations for the memory arrays **250**). In some examples, such implementations may support evaluations, configurations, or other operations of the system **200**, via contacts **234** that are accessible at a physical interface of the system, during manufacturing, assembly, validation, or other operation associated with the system **200** (e.g., before coupling with a host processor **210**, without implementing a host processor **210**, for operations independent of a host processor). Additionally, or alternatively, a logic block **230** may implement one or more aspects of a controller **215**. For example, a logic block **230** may include or operate as one or more controllers **215** and may perform operations ascribed to a controller **215**.

[0042] Each interface block **220** may be coupled with at least a respective bus **221** of the die **205**, and a respective bus **246** of a die **240**, that are configured to communicate signaling with a corresponding interface block **245** (e.g., via one or more associated signal paths). For example, the interface block **220-*a*-1** may be coupled with the interface block **245-*a*-1** via a bus **221-*a*-1** and a bus **246-*a*-1**, and the interface block **220-*a*-2** may be coupled with the interface block **245-*a*-2** via a bus **221-*a*-2** and a bus **246-*a*-2**. In some examples, a die **240** may include a bus that bypasses operational circuitry of the die **240** (e.g., bypasses interface blocks **245** of a given die **240**), such as a bus **255**. For example, the interface block **220-*a*-2** may be coupled with the interface block **245-*a*-2** of the die **240-*a*-2** via a bus **255-*a*-1** of the die **240-*a*-1**, which may bypass interface blocks **245** of the die **240-*a*-1**. Such techniques may be extended for interconnection among more than two dies **240** (e.g., for interconnection via a respective bus **255** of multiple dies **240**).

[0043] The respective signal paths of buses **221**, **246**, and **255** may be coupled with one another, from one die to another, via various arrangements of contacts at the surfaces of interfacing dies. For example, the bus **221-*a*-1** may be coupled with the bus **246-*a*-1** via a contact **222-*a*-1** of (e.g., at a surface of) the die **205** and a contact **247-*a*-1** of the die **240-*a*-1**, the bus **221-*a*-2** may be coupled with the bus **255-*a*-1** via a contact **222-*a*-2** of the die **205** and a contact **256-*a*-1** of the die **240-*a*-1**, the bus **255-*a*-1** may be coupled with the bus **246-*a*-2** via a contact **257-*a*-1** of the die **240-*a*-1** and a contact **247-*a*-2** of the die **240-*a*-2**, and so on. Although each respective bus is illustrated with a single line, coupled via singular contacts, it is to be understood that each signal path of a given bus may be associated with respective contacts to support a separate communicative coupling via each signal path of the given bus. In some examples, a bus **255** may traverse a portion of a die **240** (e.g., in an in-plane direction, along a direction different from a thickness direction, in a waterfall arrangement), which may support an arrangement of contacts **222** along a surface of a die **205** being coupled with interface blocks **245** of different dies **240** along a stack direction (e.g., via respective contacts **256** and **257** that are non-overlapping when viewed along a thickness direction).

[0044] The interconnection of interfacing contacts may be supported by various techniques. For example, in a hybrid bonding implementation, interfacing contacts may be coupled by a fusion of conductive materials (e.g., electrically conductive materials) of the interfacing contacts (e.g., without solder or other intervening material between contacts). For example, in an assembled condition, the coupling of the die **205** with the die **240-*a*-1** may include a conductive material of the contact **222-*a*-2** being fused with a conductive material of the contact **256-*a*-1**, and the coupling of the die **240-*a*-1** with the die **240-*a*-2** may include a conductive material of the contact **257-*a*-1** being fused with a conductive material of the contact **247-*a*-2**, and so on. In some examples, such coupling may include an inoperative fusion of contacts (e.g., a non-communicative coupling, a physical coupling), such as a fusion of the contact **260-*a*-1** with the contact **256-*a*-2**, neither of which are coupled with operative circuitry of the dies **240-*a*-1** or **240-*a*-2**. In some examples, such techniques may be implemented to improve coupling strength or uniformity (e.g., implementing contacts **260**, which may not be operatively coupled with an interface block **245** or an interface block **220**), or such a coupling may be a byproduct of a repetition of components that, in various configurations, may be operative or inoperative. (e.g., where, for dies **240** with a common arrangement of contacts **256** and **257**, contacts **256-*a*-1** and **257-*a*-1** provide a communicative path between the interface block **245-*a*-2** and the interface block **220-*a*-2**, but the contacts **256-*a*-2** and **257-*a*-2** do not provide a communicative path between an interface block **245** and an interface block **220**).

[0045] In some examples, a fusion of conductive materials between dies (e.g., between contacts) may be accompanied by a fusion of other materials at one or more surfaces of the interfacing dies. For example, in an assembled condition, the coupling of the die **205** with the die **240-*a*-1** may include a dielectric material **207** (e.g., an electrically non-conductive material) of the die **205** being fused with a dielectric material **242** of the die **240-*a*-1**, and the coupling of the die **240-*a*-1** with the die **240-*a*-2** may include a dielectric material **242** of the die **240-*a*-1** being fused with a dielectric material **242** of the die **240-*a*-2**. In some examples, such dielectric materials may include an oxide, a nitride, a carbide, an oxide-nitride, an oxide-carbide, or other conversion or doping of a semiconductor material of the die **205** or dies **240**, among other materials that may support such fusion. However, coupling among dies **205** and dies **240** may be implemented in accordance with other techniques, which may implement solder, adhesives, thermal interface materials, and other intervening materials.

[0046] In some examples, dies **240** may be coupled in a stack (e.g., forming a “cube” or other arrangement of dies **240**), and the stack may subsequently be coupled with a die **205**. In some examples, a respective set of one or more dies **240** may be coupled with each die **205** of multiple dies **205** as formed in a wafer (e.g., in a chip-to-wafer bonding arrangement, before cutting the wafer of dies **205**), and the dies **205** of the wafer, each coupled with their respective set of dies **240**, may be separated from one another (e.g., by cutting at least the wafer of dies **205**). In some other examples, a respective set of one or more dies **240** may be coupled with a respective die **205** after the die **205** is separated from a wafer of dies **205** (e.g., in a chip-to-chip bonding arrangement). In some other examples, a respective set of one or more wafers each including multiple dies **240** may be coupled in a stack (e.g., in a wafer-to-wafer bonding arrangement). In various examples, such techniques may be followed by separating stacks of dies **240** from the coupled wafers, or the stack of wafers having dies **240** may be coupled with another wafer including multiple dies **205** (e.g., in a second wafer-to-wafer bonding arrangement), which may be followed by separating systems **200** from the coupled wafers. In some other examples, wafer-to-wafer coupling techniques may be implemented by stacking one or more wafers of dies **240** (e.g., sequentially) over a wafer of dies

205 before separation into systems **200**, among other examples for forming systems **200**.

[0047] The buses **221**, **246**, and **255** may be configured to provide a configured signaling (e.g., a coordinated signaling, a logical signaling, modulated signaling, digital signaling) between an interface block **220** and a corresponding interface block **245**, which may involve various modulation or encoding techniques by a transmitting interface block (e.g., via a driver component of the transmitting interface block). In some examples, such signaling may be supported by (e.g., accompanied by) clock signaling communicated via the respective buses (e.g., in coordination with signal transmission). For example, the buses may be configured to convey one or more clock signals transmitted by the interface block **220** for reception by the interface block **245** (e.g., to trigger signal reception by a latch or other reception component of the interface block **245**, to support clocked operations of the interface block **245**). Additionally, or alternatively, the buses may be configured to convey one or more clock signals transmitted by the interface block **245** for reception by the interface block **220** (e.g., to trigger signal reception by a latch or other reception component of the interface block **220**, to support clocked operations of the interface block **220**). Such clock signals may be associated with the communication (e.g., unidirectional communication, bidirectional communication) of various signaling, such as control signaling, command signaling, data signaling, or any combination thereof. For example, the buses may include one or more signal paths for communications of a data bus (e.g., one or more data channels, a DQ bus, via a data interface of the interface blocks) in accordance with one or more corresponding clock signals (e.g., data clock signals), or one or more signal paths for communications of a control bus (e.g., a command/address (C/A) bus, via a command interface of the interface blocks) in accordance with one or more clock signals (e.g., control clock signals), or any combination thereof.

[0048] Interface blocks **220**, interface blocks **245**, and logic block **230** each may include circuitry (signaling circuitry, multiplexing circuitry, processing circuitry, controller circuitry) in various configurations (e.g., hardware configurations, logic configurations, software or instruction configurations) that support the functionality allocated to the respective block for accessing or otherwise operating a corresponding set of memory arrays **250**. For example, interface blocks **220** may include circuitry configured to perform a first subset of operations that support access of the memory arrays **250**, and interface blocks **245** may include circuitry configured to support a second subset of operations that support access of the memory arrays **250**. In some examples, the interface blocks **220** and **245** may support a functional split or distribution of functionality associated with a memory system controller **155**, a local memory controller **165**, or both across multiple dies (e.g., a die **205** and at least one die **240**). In some implementations, a logic block **230** may be configured to coordinate or configure aspects of the operations of the interface blocks **220**, of the interface blocks **245**, or both, and may support implementing one or more aspects of a memory system controller **155**. Such operations, or subsets of operations, may include operations performed in response to commands from the host processor **210**, or operations performed without commands from the host processor **210** (e.g., operations determined or initiated by an interface block **220**, operations determined or initiated by an interface block **245**, operations determined or initiated by a logic block **230**), or various combinations thereof.

[0049] In some implementations, the system **200** may include one or more instances of non-volatile storage (e.g., non-volatile storage **235** of a die **205**, non-volatile storage **270** of one or more dies **240**, or a combination thereof). In some examples, a logic block **230**, interface blocks **220**, interface blocks **245**, or a combination thereof may be configured to communicate signaling with one or more instances of non-volatile storage. For example, a logic block **230**, interface blocks **220**, or interface blocks **245** may be coupled with one or more instances of non-volatile storage via one or more buses (not shown), or respective contacts (not shown), where applicable, which may each include one or more signal paths operable to communicate signaling (e.g., command signaling, data signaling). In some examples, a logic block **230**, one or more interface blocks **220**, one or more interface blocks **245**, or a combination thereof may configure one or more operations based on information stored in one or more instances of non-volatile storage. Additionally, or alternatively, in some examples, a logic block **230**, one or more interface blocks **220**, one or more interface blocks **245**, or a combination thereof may write information (e.g., configuration information) to be stored in one or more instances of non-volatile storage. In some examples, such non-volatile storage may include fuses, antifuses, or other types of one-time programmable storage elements, or any combination thereof.

[0050] In some implementations, the system **200** may include one or more sensors (e.g., one or more sensors **237** of a die **205**, one or more sensors **275** of one or more dies **240**, or a combination thereof). In some implementations, a logic block **230**, interface blocks **220**, interface blocks **245**, or a combination thereof may be configured to receive one or more indications based on measurements of one or more sensors of the system **200**. For example, a logic block **230**, interface blocks **220**, or interface blocks **245** may be coupled with one or more sensors via one or more buses (not shown), or respective contacts (not shown). Such sensors may include temperature sensors, current sensors, voltage sensors, counters, and other types of sensors. In some examples, a logic block **230**, one or more interface blocks **220**, one or more interface blocks **245**, or a combination thereof may configure one or more operations based on output of the one or more sensors. For example, a logic block **230** may configure one or more operations of interface blocks **220** based on signaling (e.g., indications, data) received from the one or more sensors. Additionally, or alternatively, an interface block **220** may generate access signaling for transmitting to a corresponding interface block **245** based on one or more sensors.

[0051] In some examples, circuitry of interface blocks **220**, interface blocks **245**, or a logic block **230**, or any combination thereof may include components (e.g., transistors) formed at least in part from doped portions of a substrate of the respective die. In some examples, a substrate of a die **205** may have characteristics that are different from those of a substrate of a die **240**. Additionally, or alternatively, in some examples, transistors formed from a substrate of a die **205** may have characteristics (e.g., manufacturing characteristics, performance characteristics) that are different from transistors formed from a substrate of a die **240** (e.g., in accordance with different transistor architectures).

[0052] In some examples, the interface blocks **220** may support a layout for one or more components within the **220**. For example, the layout may include pairing components to share an access port (e.g., a command port, a data port). Further, in some examples, the layout may support interfaces for a controller **215** (e.g., a host interface **216**) that are different from interfaces for an interface block **245** (e.g., via the buses **221**). For instance, a host interface **216** may be synchronous and have separate channels for read and write operations, while an interface via buses **221** and **246** may be asynchronous and support both read and write operations with a same channel.

[0053] A die **240** may include one or more units **265** (e.g., modules) that are separated from a semiconductor wafer having a pattern (e.g., a two-dimensional pattern) of units **265**. Although each die **240** of the system **200** is illustrated with a single unit **265** (e.g., unit **265**-*a*-**1** of die **240**-*a*-**1**, unit **265**-*a*-**2** of die **240**-*a*-**2**), a die **240** in accordance with the described techniques may include any quantity of units **265**, which may be arranged in various patterns (e.g., sets of one or more units **265** along a row direction, sets of one or more units **265** along a column direction, among other patterns). Each unit **265** may include at least the circuitry of a respective interface block **245**, along with memory array(s) **250**, a bus **251**, a bus **246**, and one or more contacts **247** corresponding to the respective interface block **245**. In some examples, where applicable, each unit **265** may also include one or more buses **255**, contacts **256**, contacts **257**, or contacts **260** (e.g., associated with a respective interface block **245** of a unit **265** of a different die **240**), which may support various degrees of stackability or modularity among or via units **265** of other dies **240**.

[0054] In some examples, the interface blocks **220** may include circuitry configured to receive first access command signaling from a host processor **210** or a controller **215** (e.g., via a host interface **216**, via one or more contacts **212** from a host processor **210** or controller **215** external to a die

205), and to transmit second access command signaling to the respective (e.g., coupled) interface block **245** based on (e.g., in response to) the received first access command signaling. The interface blocks **245** may accordingly include circuitry configured to receive the second access command signaling from the respective interface block **220** and, in some examples, to access a respective set of one or more memory arrays **250** based on (e.g., in response to) the received second access command signaling. In various examples, the first access command signaling may include access commands that are associated with a type of operation (e.g., a read operation, a write operation, a refresh operation, a memory management operation), which may be associated with an indication of an address of the one or more memory arrays **250** (e.g., a logical address, a physical address). In some examples, the first access command signaling may include an indication of a logical address associated with the memory arrays **250**, and circuitry of an interface block **220** may be configured to generate the second access command signaling to indicate a physical address associated with the memory arrays **250** (e.g., a row address, a column address, using a logical-to-physical (L2P) table or other mapping or calculation functionality of the interface block **220**).

[0055] In some examples, to support write operations of the system **200**, circuitry of the interface blocks **220** may be configured to receive (e.g., from a host processor **210**, from a controller **215**, via a host interface **216**, via one or more contacts **212** from a host processor **210** or controller **215** external to a die **205**) first data signaling associated with the first access command signaling, and to transmit second data signaling (e.g., associated with second access command signaling) based on received first access command signaling and first data signaling. The interface blocks **245** may accordingly be configured to receive second data signaling, and to write data to one or more memory arrays **250** (e.g., in accordance with an indicated address associated with the first access command signaling) based on the received second access command signaling and second data signaling. In some examples, the interface blocks **220** may include an error control functionality (e.g., error detection circuitry, error correction circuitry, error correction code (ECC) logic, an ECC engine) that supports the interface blocks **220** generating the second data signaling based on performing an error control operation using the received first data signaling (e.g., detecting or correcting an error in the first data signaling, determining one or more parity bits to be conveyed in the second data signaling and written with the data).

[0056] In some examples, to support read operations of the system **200**, circuitry of the interface blocks **245** may be configured to read data from the memory arrays **250** based on received second access command signaling, and to transmit first data signaling based on the read data. The interface blocks **220** may accordingly be configured to receive first data signaling, and to transmit second data signaling (e.g., to a host processor **210**, to a controller **215**, via a host interface **216**, via one or more contacts **212** to a host processor **210** or controller **215** external to a die **205**) based on the received first data signaling. In some examples, the interface blocks **220** may include an error control functionality that supports the interface blocks **220** generating the second data signaling based on performing an error control operation using the received first data signaling (e.g., detecting or correcting an error in the first data signaling, which may include a calculation involving one or more parity bits received with the first data signaling).

[0057] In some examples, access command signaling that is transmitted by the interface blocks **220** to the interface blocks **245**, among other signaling, may be generated (e.g., based on access command signaling received from a host processor **210**, based on initiation signaling received from a host processor **210**, without receiving or otherwise independent from signaling from a host processor **210**) in accordance with various determination or generation techniques configured at the interface blocks **220** (e.g., based on a configuration for accessing memory arrays **250** that is modified at the interface blocks **220**). In some examples, such techniques may involve signaling or other coordination with a logic block **230**, a host processor **210**, one or more controllers **215**, one or more instances of non-volatile storage, one or more sensors, or any combination thereof. Such techniques may support the interface blocks **220** configuring aspects of the access operations performed on the memory arrays **250** by a respective interface block **245**, among other operations. For example, interface blocks **220** may include evaluation circuitry, access configuration circuitry, signaling circuitry, scheduling circuitry, repair circuitry, refresh circuitry, error control circuitry, adverse access circuitry, and other circuitry operable to configure operations associated with one or more dies (e.g., operations associated with accessing memory arrays **250** of the dies **240**).

[0058] In some cases, to manufacture a system **200**, semiconductor wafers including dies **240** may be stacked and coupled together, and respective stacks of dies **240** may be individually separated (e.g., cut out) from the stack of semiconductor wafers (e.g., rather than selecting individual semiconductor dies for stacking and coupling). In some examples, a semiconductor wafer including dies **205** may also be stacked and coupled with the stack of semiconductor wafers, and respective stacks of dies **240** and a die **205** that form a system **200** may be individually separated from the stack of semiconductor wafers. In some other examples, a die **205** may be stacked and coupled with a stack of dies **240**, such as after the stack of dies **240** is separated. Although wafers with a relatively high yield (e.g., of dies **240**, of dies **205**) may be selected for inclusion in a wafer stack, a non-perfect yield of dies in such wafers may be compounded in the stack of wafers such that a yield of a resulting stack of dies **240** separated from the wafer stack may be reduced. As a result, performance (e.g., latency, bandwidth, storage capacity) and yield of systems **200** may be adversely affected by such stacking.

[0059] In accordance with techniques described herein, the system **200** may be configured to support sparing techniques to improve yield of a stack of dies **240**, and thus a yield of systems **200**. For example, one or more dies **240** may be spare dies included in the stack of dies **240**. A logic block **230** of the die **205** may support sparing one or more failed components of a die **240** (e.g., a bank, a pseudo-channel, a channel, a die **240** in its entirety) with one or more components of the one or more spare dies **240** (e.g., a spare bank, a spare-pseudo-channel, a spare channel, a spare die **240** in its entirety). For instance, the logic block **230** may remap accessing of the failed components using one or more interface blocks **220** to the components of the one or more spare dies **240** using one or more other interface blocks **220**, such that accesses to an address space (e.g., an address space available to a host processor **210**, a logical address space) corresponding to the failed components may instead access the one or more components of the one or more spare dies **240**. Additionally, or alternatively, one or more dies **240** may include spare portions (e.g., one or more spare memory arrays **250**) to which the logic block **230** may remap accessing of failed components (e.g., using different interface blocks **220**). In some examples, such techniques may remap signaling of one or more host interfaces **216** from a respective first interface block **220**, or portion thereof (e.g., subset of accessed components or addresses thereof), that is associated with a failed component to a second interface block **220**, or portion thereof (e.g., subset of accessed components or addresses thereof), that is associated with a spare component (e.g., using a multiplexing or mapping component between host interfaces **216** and interface blocks **220**, not shown), which may support such sparing techniques that are transparent to a host system **105**.

[0060] FIG. **3** shows an example of an interface architecture **300** that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein. The interface architecture **300** illustrates an example of an interface block **245-*b*** (e.g., of a die **240**) coupled with an interface block **220-*b*** (e.g., of a die **205**). The interface block **245-*b*** may be communicatively coupled with the interface block **220-*b*** via one or more of a bus **301**, a bus **302**, a bus **303**, and a bus **304**, each of which may be examples of one or more signal paths of a bus **221** and a bus **246**, as well as a bus **255**, where applicable.

[0061] The interface block **245-*b*** includes a control interface **310** (e.g., a command interface), which may be configured to communicate signaling with the interface block **220-*b***. For example, the control interface **310** may include circuitry (e.g., a receiver, one or more latches) configured to receive control signaling (e.g., modulated control signaling, access command signaling, configuration signaling, address signaling, such as row address signaling or column address signaling) via the bus **301**-*a*. The control interface **310** also may include circuitry configured to receive clock signaling (e.g., clock signaling associated with the control interface **310**, clock signaling having one or more phases, such as true and complement phases, dk_t/c signaling from the interface block **220**-*b*) via the bus **302**-*a*, which the control interface **310** may use for receiving the control signaling of the bus **301**-*a* (e.g., for triggering the one or more latches). The control interface **310** may transmit (e.g., forward) the control signaling the clock signaling (e.g., for timing of other operations of the interface block **245**-*b*) to an interface controller **320**.

[0062] The interface block **245**-*b* also includes two data interfaces **330** (e.g., data interfaces **330**-*a*-**1** and **330**-*a*-**2**), which also may be configured to communicate signaling with the interface block **220**-*b*. Each data interface **330** may include corresponding buses and circuitry, the operation of which may be associated with (e.g., controlled by, coordinated with, operated based on) control signaling via the control interface **310**. Although the example of interface block **245**-*b* includes two such data interfaces **330** associated with the control interface **310** (e.g., in a “channel pair” arrangement), the described techniques for an interface block **245** may include any quantity of one or more data interfaces **330**, and associated buses and circuitry, for a given control interface **310** of the interface block **245**. Each data interface **330** may be associated with respective data path circuitry, which may include respective first-in-first-out (FIFO) and serialization/deserialization (SERDES) circuitry (e.g., FIFO/SERDES **340**), respective write/sense circuitry **350**, respective synchronization and sequencing circuitry (e.g., sync/seq logic **360**), and respective timing circuitry **370**, along with interconnecting signal paths (e.g., one or more buses). However, in some other examples, data path circuitry may be arranged in a different manner, or may include different circuitry components, which may include circuitry that is dedicated to respective data paths, or shared among data paths, or various combinations thereof. Each data interface **330** also may be associated with a respective set of one or more memory arrays **250**. In some examples, each memory array **250** may be understood to include respective addressing circuitry such as bank logic or decoders (e.g., a row decoder, a column decoder), among other array circuitry. However, in some other examples, at least a portion of such circuitry may be included in an interface block **245**.

[0063] Each data interface **330** may include circuitry (e.g., one or more latches, one or more drivers) configured to communicate (e.g., receive, transmit) data signaling (e.g., modulated data signaling, DQ signaling) via a respective bus **303**. Each data interface **330** also may include circuitry to communicate clock signaling via a respective bus **304**, which may support clock signal reception by the data interface **330** (e.g., first clock signaling associated with the data interface **330**, clock signaling having one or more phases, such as true and complement phases, DQS_t/c signaling from the interface block **220**-*b*, clock signaling associated with data reception or write operations), or clock signal transmission by the data interface **330** (e.g., second clock signaling associated with the data interface **330**, RDQS_t/c signaling to the interface block **220**-*b*, clock signaling associated with data transmission or read operations), or both. In some examples, a data interface **330**, a bus **303**, or a combination of a bus **303** and a bus **304**, may be associated with a “pseudo-channel,” and multiple pseudo-channels may be associated with a same control interface **310** or a same control bus (e.g., a bus **301**, a combination of a bus **301** and a bus **302**). Each data interface **330** may transmit clock signaling (e.g., received clock signaling, DQS_t/c signaling) to sync/seq logic **360** via a respective bus (e.g., for timing of other operations of the interface block **245**-*b*).

[0064] The interface controller **320** may support various control or configuration functionality of the interface block **245**-*b* for accessing or otherwise managing operations of the coupled memory arrays **250**. For example, the interface controller **320** may support access command coordination or configuration, latency or timing compensation, access command buffering (e.g., in accordance with a FIFO or other organizational scheme), mode registers or logic for configuration settings, or test functionality, among other functions or combinations thereof. For each data path of the interface block **245** (e.g., associated with a respective data interface **330**), the interface controller **320** may be configured to transmit signaling to the respective memory arrays **250** via a bus **321** (e.g., address signaling, such as a row address or row activation signaling). For each data path of the interface block **245**, the interface controller **320** may communicate signaling (e.g., timing signaling, which may be based on clock signaling received from the control interface **310**, configuration signaling) with respective timing circuitries **370** and sync/seq logic **360** via respective buses.

[0065] For each data path, the respective timing circuitry **370** may support timing of various operations (e.g., activations, coupling operations, signal latching, signal driving) relative to timing signaling received from the interface controller **320**. For example, timing circuitry **370** may include a timing chain (e.g., a global column timing chain) configured to generate one or more clock signals or other initiation signals for controlling operations of the respective data path, and such signaling may include transitions (e.g., rising edge transitions, falling edge transitions, on/off transitions) that are offset from, at a different rate from, or otherwise different from transitions of signaling from the interface controller **320** to support a given operation or combination of operations. For example, timing circuitry **370** may be configured to transmit signaling (e.g., column selection signaling, column address signaling) to the respective memory arrays **250**, to transmit signaling to the respective write/sense circuitry **350** (e.g., latch or driver timing signaling), and to transmit signaling to the respective sync/seq logic (e.g., timing signaling).

[0066] For each data path, the respective FIFO/SERDES **340** may be configured to convert between data signaling of a first bus width (e.g., a relatively wide bus width, a data read/write (DRW) bus, a bus for communications with write/sense circuitry **350** having a relatively larger quantity of signal paths) and a second bus width (e.g., a relatively narrow bus width, a bus for communications with a data interface **330** having a relatively smaller quantity of signal paths). In some examples, such a conversion may be accompanied by changing a rate of signaling between signaling from the data interface **330** and the write/sense circuitry **350** (e.g., to maintain a given throughput). In various examples, the FIFO/SERDES **340** may receive data signaling from the data interface **330** and transmit data signaling to the write/sense circuitry **350** (e.g., to support a write operation), or may receive data signaling from the sense circuitry **350** and transmit data signaling to the data interface **330** (e.g., to support a read operation). In some examples (e.g., to support a read operation), the FIFO/SERDES **340** may be configured to transmit clock signaling (e.g., RDQS_t/c signaling) to the data interface **330**, which may be forwarded to the interface block **220-*b***.

[0067] The timing or other synchronization of operations performed by the FIFO/SERDES **340** may be supported by one or more clock signals, among other signaling, received from the respective sync/seq logic **360**. For example, the sync/seq logic **360** may generate or otherwise coordinate clock signaling to support the different rates of signaling of different buses (e.g., based on received clock signaling). Additionally, or alternatively, the FIFO/SERDES **340** may operate in a direction (e.g., for data transmission to a data interface **330**, for data reception from a data interface **330**) or other mode based on configuration signaling received from the sync/seq logic **360**.

[0068] For each data path, the respective write/sense circuitry **350** may be configured to support the accessing (e.g., data signaling, write signaling, read signaling) of the respective set of one or more memory arrays **250**. For example, the write/sense circuitry **350** may be coupled with the memory arrays **250** via a bus (e.g., a global input/output (GIO) bus), which may include respective signal paths associated with each memory array **250**, or may include signal paths that are shared for all of the memory arrays **250** of the set, in which case the memory array circuitry may include multiplexing circuitry operable to couple the bus with a selected one of the memory arrays **250**. In some examples, a bus between the write/sense circuitry and the set of one or more memory arrays **250** may include a same quantity of signal paths as a bus between the write/sense circuitry **350** and the FIFO/SERDES **340** (e.g., for signaling GIO[287:0]) or a same quantity of signal paths as a quantity of columns in each memory array **250**. In some other examples, the memory arrays **250** may include a quantity of columns that is an integer multiple of the quantity of signal paths of the bus, in which case the memory array circuitry (e.g., each memory array **250**) may include decoding circuitry operable to couple a subset of columns of memory cells, or associated circuitry, with the bus.

[0069] To support write operations, the write/sense circuitry **350** may be configured to drive signaling that is operable to write one or more logic states to memory cells of the memory arrays **250** (e.g., based on received data, based on received timing signaling, based on data signaling received via a bus **303** and on control signaling received via a bus **301**-*a*). In some examples, such signaling may be transmitted to supporting circuitry of or otherwise associated with the memory arrays **250** (e.g., as an output of signals corresponding to logic states to be written), such as sense amplifier circuitry, voltage sources, current sources, or other driver circuitry operable to apply a bias across a storage element of the memory cells (e.g., across a capacitor, across a ferroelectric capacitor), or apply a charge, a current, or other signaling to a storage element of the memory cells (e.g., to apply a current to a chalcogenide or other configurable memory material, to apply a charge to a gate of a NAND memory cell), among other examples.

[0070] To support read operations, the write/sense circuitry **350** may be configured to receive signaling that the write/sense circuitry **350** may further amplify for communication through the interface block **245**-*b*. For example, the write/sense circuitry **350** may be configured to receive signaling corresponding to logic states read from the memory arrays **250**, but at a relatively low driver strength (e.g., relatively 'analog' signaling, which may be associated with a relatively low drive strength of sense amplifiers of the memory arrays **250**). The write/sense circuitry **350** may thus include further sense amplification (e.g., a data sense amplifier (DSA) between signal paths between the write/sense circuitry and the set of one or more memory arrays and respective signal paths between the write/sense circuitry and the FIFO/SERDES), which each may have a relatively high drive strength (e.g., for driving relatively 'digital' signaling).

[0071] The features of the interface architecture **300** may be duplicated in various quantities and arrangements to support a semiconductor system having multiple dies, such as various examples of a system **200**. In an example implementation, each die **240** may be configured with 64 instances of the interface block **245**-*b*, which may support a data signaling width of 9,216 signal paths for each die **240** (e.g., where each bus **303** of a channel pair is associated with 72 signal paths). For a system **200** having a stack of eight dies **240** coupled with a die **205**, the die **205** may thus be configured with 512 instances of the interface block **220**-*b*, thereby supporting an overall data signaling width of 73,738 signal paths for the system **200**. However, in other implementations, dies **205** and dies **240** may be configured with different quantities of interface blocks **220** and **245**, respectively, and a system **200** may be configured with different quantities of dies **240** per die **205**.

[0072] In accordance with techniques described herein, a system that implements the interface architecture **300** (e.g., a system **200**) may support sparing techniques to improve yield of a stack of dies **240**, and thus a yield of related stacked semiconductor systems. For example, one or more dies **240** may be spare dies included in the stack of dies **240**. A logic block **230** of the die **205** may support sparing one or more failed components (e.g., a bank, a pseudo-channel, a channel, a die **240**) of the dies **240** with one or more components (e.g., a bank, a pseudo-channel, a channel, a die **240**) of the one or more spare dies **240**. For instance, the logic block **230** may remap accessing of the failed components using one or more interface blocks **220** to the components of the one or more spare dies **240** using one or more other interface blocks **220** such that accesses to an address space (e.g., available to a host processor **210**) corresponding to the failed components may instead access the one or more components of the one or more spare dies **240**. Additionally, or alternatively, one or more dies **240** may include spare portions (e.g., one or more spare memory arrays **250**) to which the logic block **230** may remap accessing of failed components (e.g., using different interface blocks **220**).

[0073] In some examples, sparing techniques may include a remapping of at least a portion of interface blocks **220**-*b*, or interface blocks **245**-*b*, or both for a given host interface **216** (e.g., isolating an interface block **220**-*b* associated with a component failure from a host interface **216** and coupling an interface block **220**-*b* associated with a spare component with the respective host interface **216**). To support die-level sparing, for example, a logic block **230** may remap (e.g., for respective host interfaces **216**) each instance of an interface block **245**-*b* in a first die **240** (e.g., and corresponding interface block **220**-*b*) to a respective instance of an interface block **245**-*b* in a second die **240** (e.g., and corresponding interface block **220**-*b*). To support channel-level sparing, the logic block **230** may remap a first instance of an interface block **245**-*b* (e.g., in a first die **240**) and corresponding interface block **220**-*b* to a second instance of an interface block **245**-*b* (e.g., in the first die **240**, in a second die **240**) and corresponding interface block **220**-*b*. To support pseudo-channel level sparing, the logic block **230** may remap a first pseudo-channel (e.g., a bus **303**-*a* and a bus **304**-*a*, a data channel) associated with a first instance of an interface block **245**-*b* (e.g., in a first die **240**) and, in some examples, a first instance of an interface block **220**-*b* to a second pseudo-channel associated with a second instance of an interface block **245**-*b* (e.g., in the first die **240**, in a second die **240**) and, in some examples, a second instance of an interface block **220**-*b*. To support bank-level sparing, the logic block **230** may remap a first bank (e.g., a memory array **250**-*b*, a bank of a memory array **250**-*b*) associated with a first instance of an interface block **245**-*b* (e.g., in a first die **240**) and, in some examples, a first instance of an interface block **220**-*b* to a second bank associated with a second instance of an interface block **245**-*b* (e.g., in the first die **240**, in a second die **240**) and, in some examples, a second instance of an interface block **220**-*b*.

[0074] FIG. **4** shows an example of a manufacturing process **400** that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein. For example, FIG. **4** may illustrate aspects of a sequence of manufacturing operations for fabricating a stack **425** (e.g., a stack of dies), which may be an example of at least a portion of a system **200**. Operations illustrated in and described with reference to FIG. **4** may be performed by a manufacturing system, such as a semiconductor fabrication system configured to perform additive operations such as deposition or bonding, subtractive operations such as etching, trenching, planarizing, or polishing, and supporting operations such as masking, patterning, photolithography, coupling, stacking, cutting, or aligning, among other operations that support the described techniques. In some examples, operations performed by such a manufacturing system may be supported by a process controller or its components as described herein.

[0075] A first set of one or more manufacturing operations may include the formation of dies **240**-*b* on a wafer **405**. For example, the wafer **405** may be a semiconductor wafer on which a set of dies **240**-*b* are formed, such as via various additive operations, subtractive operations, supporting operations, doping operations, or a combination thereof, among other types of operations.

[0076] To manufacture a system **200**, such as a 3D stacked memory system, coupling (e.g., bonding) of dies **240**-*b* in a stack may occur at the wafer level. For example, a second set of one or more manufacturing operations may include the coupling of a set of wafers **405** into a stack **410** (e.g., a stack of wafers). In some examples, coupling the wafers **405** may include fusing (e.g., bonding) respective conductive materials of respective contacts (e.g., contacts **247**, contacts **256**, contacts **257**, contacts **260**) of dies **240**-*b*. As such, the stack **410** may include multiple stacks of dies **240**-*b* that are stacked along a direction (e.g., a direction along which the wafers **405** are stacked).

[0077] In some examples, a third set of one or more manufacturing operations may include coupling a wafer **415** with the stack **410** to form a stack **420** (e.g., a stack of wafers). The wafer **415** may be a semiconductor wafer on which a set of dies **205**-*b* are formed, such as via various additive operations, subtractive operations, supporting operations, doping operations, or a combination thereof, among other types of operations. The third set of manufacturing operations may include fusing (e.g., bonding) respective conductive materials of respective contacts (e.g., contacts **222**) of the set of dies **205**-*b* with respective conductive materials of respective contacts (e.g., contacts **247**, contacts **256**) of the dies **240**-*b*. For example, respective contacts **222** of the set of dies **205-*b*** may be coupled (e.g., fused, bonded) with contacts of a respective bottom die **240-*b*** of each stack of dies **240-*b*** while the set of dies **205-*b*** are included in (e.g., a part of) the wafer **415** and the stacks of dies **240-*b*** are included in (e.g., a part of) the wafers **405**.

[0078] In some examples, a fourth set of one or more manufacturing operations may include separating a stack **425** from the stack **420**. For example, the fourth set of one or more manufacturing operations may include cutting respective stacks **425** from the stack **420**, where each stack **425** includes a respective die **205-*b*** of the wafer **415** and a respective stack of dies **240-*b*** coupled with the respective die **205-*b***. Each stack **425** may be an example of a respective system **200**.

[0079] In some examples, the coupling of a die **205-*b*** with a stack of dies **240-*b*** to form the stack **425** may occur after the stack of dies **240-*b*** are separated from the stack **410**. For example, the third set of one or more manufacturing operations may include cutting respective stacks of dies **240-*b*** from the stack **410**. The fourth set of one or more manufacturing operations may include separating (e.g., cutting) a respective die **205-*b*** from the wafer **415** and coupling the respective die **205-*b*** with a respective stack of dies **240-*b*** to form a respective stack **425**. Although some techniques are described in the context of wafer stacking, the sparing techniques described herein may be implemented in stacks **425** formed by other techniques, such as those that involve directly stacking dies **240** and **205** (e.g., after one or more dies **240** or **205** are separated individually from one or more wafers).

[0080] The manufacturing of stacks **425** may be configured support sparing techniques as described herein. For example, a set of wafers **405** that are coupled together to form a stack **410** may include one or more additional (e.g., spare) wafers **405** such that each stack of dies **240-*b*** that are separated from the stack **410** and included in a respective stack **425** may include one or more additional (e.g., spare) dies **240-*b***. For example, a stack **425** may include a die **240-*b*-1**, which may be an example of a spare array die including components (e.g., banks, channels, pseudo-channels, the die **240-*b*-1** in its entirety) to which accessing may be remapped. In some examples, one or more dies **240-*b*** may additionally, or alternatively, include additional (e.g., spare) portions including spare components (e.g., banks, channels, pseudo-channels, memory arrays) to which accessing may be remapped.

[0081] In some implementations, one or more functions of each die **240-*b*** may be evaluated at various stages of the manufacturing process **400** to determine whether to perform sparing associated with the die **240-*b***. In some examples, a function of each die **240-*b*** may be evaluated (e.g., by a manufacturing system, by a tester) before the stack **410** is formed. For instance, coupling the set of wafers **405** to form the stack **410** may occur after the function of each die **240-*b*** of each wafer **405** has been evaluated. Additionally, or alternatively, a function of each die **240-*b*** may be evaluated after the stack **410** is formed (e.g., before the stack **420** is formed, before a die **205-*b*** is coupled with a respective stack **425**). For example, a set of wafers **405** may be coupled together to form a stack **410** after which the function of each die **240-*b*** may be evaluated. In some examples, a function of each die **240-*b*** may be evaluated after a respective stack of dies **240-*b*** is separated from a stack **410** and before a respective die **205-*b*** is coupled with the respective stack of dies **240-*b***. Additionally, or alternatively, a function of each die **240-*b*** may be evaluated after a stack **425** is formed. For example, a stack of dies **240-*b*** may be coupled with a die **205-*b***, including as part of a stack **420** or as part of a stack **425**, and a function of each die **240-*b*** of the stack of dies **240-*b*** may be evaluated, for example, using circuitry of a die **205-*b*** (e.g., interface blocks **220** of the die **205-*b***).

[0082] Evaluating the function of each die **240-*b*** may support the detection of one or more errors associated with a given die **240-*b*** to determine whether to perform sparing for the die **240-*b***. For example, evaluating the function of a die **240-*b*** may include performing one or more operations (e.g., access operations) to determine whether components of the die **240-*b*** (e.g., of interface blocks **245**, of memory arrays **250**, of buses **251**, of buses **255**, of buses **246**, of contacts or couplings thereof) are functioning properly and whether an error (e.g., failure) associated with a given component exists. In some examples, evaluating the function of a die **240**-*b* may be relatively more granular after a stack **425** is formed. For example, circuitry of interface blocks **220** of a die **205**-*b* of a stack **425** may be used to evaluate respective functions of dies **240**-*b* of the stack **425** at a finer granularity than circuitry of a tester used to evaluate the respective functions before the formation of the stack **425**. As such, in some cases, relatively more types of errors may be detected as part of an evaluation after the formation of a stack **425** than as part of an evaluation before the formation of the stack **425**.

[0083] In some examples, an error associated with a die **240**-*b* may be detected during operation of a stack **425** (e.g., during evaluation operations, during operations in an assembled or integrated condition, in a deployment of a stack **425**). For example, errors may occur over time due to wear out or adverse operation (e.g., over-voltage operations, over-temperature operations), among other factors. In some examples, an error may occur as part of an attempted access of a memory array **250** of a die **240**-*b* using an interface block **220** of a die **205**-*b*. In some examples, the error may be detected by the interface block **220** based on a failure of the attempted access. In some examples, the interface block **220** may indicate (e.g., transmit an indication of) the error to a logic block **230** of the die **205**-*b*.

[0084] In some examples, non-volatile storage of one or more dies **240**-*b* or of a die **205**-*b* may be set to indicate the error, and a logic block **230** may detect (e.g., identify) the error for the purposes of sparing based on accessing the non-volatile storage. For example, non-volatile storage **270** of a first die **240**-*b* at which an error occurs may be set to indicate the error. Additionally, or alternatively, non-volatile storage **270** of a second die **240**-*b* to which accessing is remapped may be set to indicate the error associated with the first die **240**-*b*. Additionally, or alternatively, non-volatile storage **235** of a die **205**-*b* may be set to indicate the error associated with first die **240**-*b* (e.g., and to indicate the second die **240**-*b* to which accessing is remapped). In some examples, to set a non-volatile storage **270** or non-volatile storage **235**, one-time programmable storage elements of the non-volatile storage may be set. For example, one or more fuses or antifuses of the non-volatile storage may be set (e.g., blown) to indicate the error. In some cases, a tester performing the evaluation of a die **240**-*b* may set a non-volatile storage **270** of the die **240**-*b*. In some cases, circuitry of a logic die **205**-*b* (e.g., an interface block **220**, a logic block **230**) may set (e.g., cause the setting of) a non-volatile storage **270** of a die **240**-*b* or a non-volatile storage **235** of the logic die **205**-*b*.

[0085] Sparing may be performed based on the detection (e.g., identification) of an error associated with a die **240**-*b*. For example, a logic block **230** of a die **205**-*b* may remap accessing (e.g., via one or more host interfaces **216**) of a stack of dies **240**-*b* based on (e.g., in response to) detecting the error, such as by accessing non-volatile storage **270** or **235** or based on a failed attempted access of one or more memory arrays **250** of the stack of dies **240**-*b*. A logic block **230** may support remapping accessing of the stack of dies **240**-*b* according to various levels of granularity, such as based on a type of the detected error. For example, a logic block **230** may support remapping at a die level of granularity, the channel level of granularity, a pseudo-channel level of granularity, a bank level of granularity, or a combination thereof.

[0086] By supporting sparing based on error detection, the yield of stacks **425** (e.g., systems **200**, 3D stacked memory systems) may be increased, for example, by supporting the replacement of failed components with spare components. As such, performance of stacks **425**, including bandwidth, storage capacity, latency, may be improved. Additionally, a time-to-market of products that implement the stack **425**. For example, products released to the market may be associated with yield constraints, and improving yield via the sparing techniques described herein may enable such constraints to be satisfied without improving manufacturing techniques to increase the yield of individual semiconductor wafers, which may involve a delay before such improvements are realized. Additionally, the sparing techniques described herein may reduce costs as a final yield of a stack of dies **240**-*b* may be unknown until after it is coupled with a die **205**-*b* (e.g., due to finer granularity at which the function of the dies **240**-*b* may be tested after coupling with the die **205**-*b*). Accordingly, if a yield fails to meet yield constraints, the entire stack **425** may be discarded. Thus, improving yield of stacks of dies **240**-*b* using the sparing techniques described herein may increase the likelihood that yield constraints are met, thereby reducing costs of manufacturing stacks **425**.

[0087] FIGS. **5**A, **5**B, **5**C, and **5**D show examples of sparing diagrams **500** that support sparing techniques in stacked memory architectures in accordance with examples as disclosed herein. The sparing diagrams **500** illustrate examples of remapping operations that may be performed to support sparing within a stack **425** (e.g., a system **200**, a 3D stacked memory system) in accordance with various levels of granularity.

[0088] FIG. **5**A shows a sparing diagram **500**-*a* that supports die-level sparing. For example, the sparing diagram **500**-*a* depicts a die **205**-*c*-**1** coupled with a set of dies **240**-*c* stacked along a direction (e.g., a z-direction, a direction perpendicular to the die **205**-*c*-**1**), which may collectively be an example of a stack **425**. An error associated with a die **240**-*c*-**1** of the set of dies **240**-*c* may be detected as described herein. Based on the detection of the error, a logic block **230** of the die **205**-*c*-**1** may remap access of the die **240**-*c*-**1** to a die **240**-*c*-**2** of the set of dies **240**-*c*. For example, an address space available to (e.g., accessible by) a host system (e.g., a host system **105**, a host processor **210**, such as of the die **205**-*c*-**1** or included in another die coupled with the die **205**-*c*-**1**) via a set of host interfaces **216** of the die **205**-*c*-**1** may include a portion of addresses that may be used by the host system to access the die **240**-*c*-**1** (e.g., memory arrays **250** of the die **240**-*c*-**1**). The logic block **230** may remap the portion of addresses such that use of addresses within the portion by the host system (e.g., via a subset of host interfaces **216** of the die **205**-*c*-**1**) may instead access the die **240**-*c*-**2** (e.g., memory arrays **250** of the die **240**-*c*-**2**). In some examples, the die **240**-*c*-**2** may be a spare die of the set of dies **240**-*c*.

[0089] The die **240**-*c*-**1** may include interface blocks **245** via which the memory arrays **250** of the die **240**-*c*-**1** may be accessed. To support remapping at the die level of granularity (e.g., the entirety of the portion of the address space corresponding to the die **240**-*c*-**1**), the logic block **230** may remap access (e.g., via the subset of host interfaces **216** that correspond to the die **240**-*c*-**1**) from all of the interface blocks **245** of the die **240**-*c*-**1** to respective interface blocks **245** of the die **240**-*c*-**2**. For example, the portion of the address space corresponding to the die **240**-*c*-**1** may be remapped to the interface blocks **245** of the die **240**-*c*-**2**. In some examples, a first set of interface blocks **220** of the die **205**-*c*-**1** may be coupled with the interface blocks **245** of the die **240**-*c*-**1**. To support the remapping, the logic block **230** may remap access (e.g., remap coupling with the subset of host interfaces **216**) from the first set of interface blocks **220** to a second set of interface blocks **220** of the die **205**-*c*-**1** that are coupled with the interface blocks **245** of the die **240**-*c*-**2**. For example, the portion of the address space corresponding to the die **240**-*c*-**1** may be remapped to the second set of interface blocks **220** such that accesses of addresses within the remapped portion of the address space may be supported by respective interface blocks **220** of the second set of interface blocks **220**. In some examples, the second set of interface blocks **220** may be referred to as spare interface blocks **220** of the die **205**-*c*-**1** (e.g., based on being coupled with interface blocks **245** of a spare die **240**-*c*).

[0090] In some examples, one or more spare dies **240** (e.g., for die-level sparing, for channel-level, sparing, for pseudo-channel-level sparing, for bank-level sparing) in accordance with the described techniques may be located at various positions (e.g., stack positions, positions along the z-direction) in a stack **425** to support various operating characteristics of the system. For example, one or more spare dies **240** may be located at the bottom of a stack **425** (e.g., spare dies **240** of a stack **425** that are located nearest to the die **205** along the z-direction, such as die **240**-*c*-**2** being adjacent to die **205**-*c*-**1**, one or more spare dies **240** that are between a die **205** and one or more dies **240** allocated to nominal or initial operations). In some examples, positioning one or more spare dies **240** at the bottom of a stack **425** may provide thermal benefits to the system. For example, dies

240-*c* located relatively nearer to the die **205**-*c*-**1** along the z-direction may be relatively hotter than dies **240**-*c* located further from the die **205**-*c*-**1**, which may be due to a relatively high power density associated with the die **205**-*c*-**1**, or having an otherwise unfavorable thermal path for heat rejection, among other considerations. In some cases, relatively higher temperatures may be associated with a relatively higher likelihood of errors or otherwise degraded performance of a die **240**. Accordingly, by positioning a spare die **240** relatively nearer to a die **205**, adverse thermal characteristics of the spare die **205** may be avoided in nominal (e.g., initial, as-built) configurations, and may be implicated or addressed if the spare die **240**, or some portion thereof, is implemented (e.g., by a logic block **230**). Further, dies **240** of a stack **425** that are configured for nominal operations may be relatively cooler than one or more spare dies by positioning them relatively farther from a die **205**, thereby improving nominal performance of the system (e.g., improving performance of the dies **240** for nominal operations), reducing a likelihood of errors that may involve resolution by way of sparing, or both.

[0091] In some other examples, a spare die **240** may be located at another position within a stack **425**. For example, a spare die **240** may be located at the top of a stack **425** (e.g., above other dies **240**-*c* along the z-direction). In some examples, such an implementation may be associated with relatively shorter or otherwise less-complex electrical routing (e.g., a shorter distance along the z-direction, fewer interconnections between dies), which may be associated with relatively favorable signaling characteristics in a nominal configuration, or a relatively higher nominal yield (e.g., due to relatively fewer die-to-die connections implicated for the nominal configuration, and a reduced likelihood of requiring sparing). In some other examples, locations of spare dies **240** may not be pre-determined, and spare dies **240** may allocated at various positions based on evaluations of the dies **240** themselves, or evaluations of the dies **240** after stacking, or various combinations thereof.

[0092] FIG. **5**B shows an example of a sparing diagram **500**-*b* that supports channel-level sparing. For example, the sparing diagram **500**-*b* depicts a die **205**-*c*-**2** coupled with a set of dies **240**-*c* stacked along a direction (e.g., a z-direction), which may be an example of a stack **425**. An error associated with a die **240**-*c*-**3** of the set of dies **240**-*c* may be detected as described herein. For example, each die **240**-*c* may include a set of channels **505**-*a* (e.g., 64 channels, associated with 64 units **265**) via which information may be communicated between an interface block **220** of the die **205**-*c*-**2** and an interface block **245** of the die **240**-*c*. For instance, each channel **505**-*a* may include a command channel (e.g., a bus **301** and a bus **302**) and a set of data channels (e.g., each data channel including a bus **303** and a bus **304**) via which the interface block **220** and the interface block **245** may communicate signaling. In some examples, each channel **505**-*a* may include one command channel and two data channels (e.g., a channel pair, two pseudo-channels). In some examples, groups of channels **505**-*a* within the stacked dies **240**-*c* may be arranged (e.g., organized) according to channel sets **510**-*a*. For example, a channel set **510**-*a* may include a respective channel **505**-*a* for each unit **265** across the set of dies **240**-*c*, for which signaling may be passed through buses **255** of units **265** that are stacked along the z-direction. In some examples, an interface block **220** of the die **205**-*c*-**2** may be associated with a respective channel **505**-*a*, or a channel set **510**-*a*. For example, the die **205**-*c*-**2** may include a respective interface block **220** for each channel **505**-*a* of the set of dies **240**-*c*.

[0093] In the example of FIG. **5**B, an error associated with a channel **505**-*a*-**1** of the die **240**-*c*-**3** may be detected. Based on the detection of the error, a logic block **230** of the die **205**-*c*-**2** may remap access of the channel **505**-*a*-**1** (e.g., of a unit **265** corresponding to the channel **505**-*a*-**1**) to another channel **505**-*a* of the set of dies **240**-*c* (e.g., to another unit **265**). For example, the logic block **230** may remap access of the channel **505**-*a*-**1** to a channel **505**-*a*-**2** of a die **240**-*c*-**4** of the set of dies **240**-*c*, which may be included in a channel set **510**-*a*-**1** that includes the channels **505**-*a*-**1** and **505**-*a*-**2**. In some other examples, the logic block **230** may remap access of the channel **505**-*a*-**1** to another channel **505**-*a* of the die **240**-*c*-**4** that is not included in the channel set **510**-*a*-**1** (e.g., included in another channel set **510**-*a*). In some other examples, the die **240**-*c*-**3** may include one or more spare channels **505**-*a*, and the logic block **230** may remap access of the channel **505**-*a*-**1** to a spare channel **505**-*a* of the die **240**-*c*-**3**. In some examples, the die **240**-*c*-**4** may be a spare die of the set of dies **240**-*c*, and the channels **505**-*a* of the die **240**-*c*-**4** may be spare channels **505**-*a*.

[0094] To support the remapping, the logic block **230** may remap a portion of an address space available to a host system that corresponds to the channel **505**-*a*-**1**. For example, the logic block **230** may remap the portion of addresses such that use of addresses within the portion by the host system may instead access the channel **505**-*a*-**2** (e.g., memory arrays **250** of a unit **265** accessible via the channel **505**-*a*-**2**) or some other channel **505**-*a* to which the logic block **230** remaps access of the channel **505**-*a*-**1**. To remap the portion of the address space, the logic block **230** may remap an interface block **245** of the die **240**-*c*-**3** associated with the channel **505**-*a*-**1** to an interface block **245** of the die **240**-*c*-**4** associated with the channel **505**-*a*-**2** or to an interface block **245** associated with some other channel **505**-*a* to which access of the channel **505**-*a*-**1** is remapped (e.g., another interface block **245** of the die **240**-*c*-**3** associated with a spare channel **505**-*a* of the die **240**-*c*-**3**).

[0095] Additionally, or alternatively, the logic block **230** may remap access from a first interface block **220** of the die **205**-*c*-**2** associated with the channel **505**-*a*-**1** to a second interface block **220** of the die **205**-*c*-**2** associated with the channel **505**-*a*-**2** or some other channel **505**-*a* to which access of the channel **505**-*a*-**1** is remapped. For example, the portion of the address space corresponding to the channel **505**-*a*-**1** may be remapped to the second interface block **220** such that accesses of addresses within the remapped portion of the address space may be supported by the second interface block **220**. In some examples, the second interface block **220** may be referred to as a spare interface block **220** (e.g., based on being coupled with an interface block **245** associated with a spare channel **505**-*a*).

[0096] In some examples, a die **240** that includes a channel **505** to which a portion of an address space is remapped (e.g., a channel **505** of a spare die **240**) may be located at various positions within a stack **425**. In some cases, a spare die **240** may be located at a bottom of the stack **425**, which may provide thermal benefits to other dies **240** of the stack **425**. For example, dies **240**-*c* located nearer to the die **205**-*c*-**2** along the z-direction may be relatively hotter than dies **240**-*c* located further from the die **205**-*c*-**2** due to a high power density associated with the die **205**-*c*-**2**. By positioning a spare die **240**-*c* nearer to the die **205**-*c*-**2**, other dies **240**-*c* of the stack **425** may be relatively cooler, thereby improving performance of a nominal configuration, or decreasing the likelihood of errors that may result in the logic block **230** performing the remapping, or both.

[0097] FIG. **5**C shows an example of a sparing diagram **500**-*c* that supports pseudo-channel-level sparing. For example, the sparing diagram **500**-*c* depicts a die **205**-*c*-**3** coupled with a set of dies **240**-*c* stacked along a direction (e.g., a z-direction), which may be an example of a stack **425**. An error associated with one or more dies **240**-*c* of the set of dies **240**-*c* may be detected as described herein. For example, each die **240**-*c* may include a set of channels **505**-*a* that each include a command channel (e.g., a bus **301**-*a* and a bus **302**-*a*) and a set of data channels (e.g., each data channel including a bus **303**-*a* and a bus **304**-*a*) via which a respective interface block **220** and interface block **245** may communicate signaling. Each channel **505**-*a* may include a set of pseudo-channels **515**, each of which may be associated with a respective data interface **330**. For example, a command channel of a channel **505**-*a* (e.g., associated with a control interface **310**) may be shared by (e.g., common to) the data channels of the channel **505**-*a*. That is, command signaling associated with communicating data via each of the data channels of the channel **505**-*a* may be communicated via the shared command channel. A pseudo-channel **515** may include a respective data channel of a channel **505**-*a* and the shared command channel. In the example of FIG. **5**C, each channel **505**-*a* may include one command channel and two data channels and may thus include two pseudo-channels **515**. In some examples, the die **205**-*c*-**3** may include a respective interface block **220** for each channel **505**-*a* of the set of dies **240**-*c*. Accordingly, each interface block **220** may be for (e.g., communicate signaling via) a set of pseudo-channels **515** included in a corresponding channel **505**-*a*.

[0098] In the example of FIG. 5C, one or more errors associated with one or more pseudo-channels **515** may be detected. For example, an error associated with a pseudo-channel **515**-*a*-**1** of a die **240**-*c*-**5** of the set of dies **240**-*c* may be detected, or an error associated with a pseudo-channel **515**-*a*-**2** of a die **240**-*c*-**6** of the set of dies **240**-*c* may be detected, or both. Based on the detection of the one or more errors, a logic block **230** of the die **205**-*c*-**3** may remap access of the pseudo-channels **515**-*a*-**1** or **515**-*a*-**2** to other pseudo-channels **515**-*a* of the set of dies **240**-*c*. For example, the logic block **230** may remap access of the pseudo-channel **515**-*a*-**1** to a pseudo-channel **515**-*a*-**3** of a die **240**-*c*-**7** of the set of dies **240**-*c*, or remap access of the pseudo-channel **515**-*a*-**2** to a pseudo-channel **515**-*a*-**4** of the die **240**-*c*-**7**, or both. In some examples, the pseudo-channels **515**-*a* to which the accessing is remapped may be included in a same channel set **510**-*a* (e.g., a channel set **510**-*a*-**2**) that includes the pseudo-channels **515**-*a*-**1** and **515**-*a*-**2**. In some other examples, the logic block **230** may remap access of the pseudo-channels **515**-*a*-**1** and pseudo-channels **515**-*a*-**2** to other pseudo-channels **515**-*a* of the die **240**-*c*-**7** that are not included in the channel set **510**-*a*-**2** (e.g., included in another channel set **510**-*a*). In some other examples, in some cases, the dies **240**-*c*-**5** and/or **240**-*c*-**6** may include one or more spare channels **505**-*a*, and the logic block **230** may remap access of the pseudo-channels **515**-*a*-**1** and pseudo-channels **515**-*a*-**2** to other pseudo-channels **515**-*a* of the dies **240**-*c*-**5** and **240**-*c*-**6**, respectively (e.g., spare pseudo-channels **515**-*a* included in a spare channel **505**-*a*). In some examples, the die **240**-*c*-**7** may be a spare die of the set of dies **240**-*c*, and the pseudo-channels **515**-*a* of the die **240**-*c*-**7** may be spare pseudo-channels **515**-*a*.
[0099] In some examples, remapping access from the pseudo-channels **515**-*a*-**1** and **515**-*a*-**2** may include remapping access from the corresponding data channels of the pseudo-channels **515**-*a*-**1** and **515**-*a*-**2** to the data channels of the pseudo-channels **515**-*a* to which the pseudo-channels **515**-*a*-**1** and **515**-*a*-**2** are remapped (e.g., pseudo-channels **515**-*a*-**3** and **515**-*a*-**4**).
[0100] To support the remapping, the logic block **230** may remap a portion of an address space available to a host system that corresponds to the pseudo-channels **515**-*a*-**1** and **515**-*a*-**2**. For example, the logic block **230** may remap the portion of addresses such that use of addresses within the portion by the host system may instead access the pseudo-channels **515**-*a*-**3** and **515**-*a*-**4**, respectively (e.g., memory arrays **250** accessible via the pseudo-channels **515**-*a*-**3** and **515**-*a*-**4**) or some other pseudo-channels **515**-*a* to which the logic block **230** remaps access of the pseudo-channels **515**-*a*-**1** and **515**-*a*-**2**. Due to pseudo-channels **515**-*a* sharing a command channel with at least one other pseudo-channel **515**-*a*, in some examples, the communication of shared command signaling via remapped pseudo-channels **515**-*a* may be coordinated. For example, the logic block **230** may configure a first interface block **220** associated with the pseudo-channel **515**-*a*-**1** and/or a second interface block **220** associated with the pseudo-channel **515**-*a*-**3** to communicate shared command signaling via a command channel of a first channel **505**-*a* that includes the pseudo-channel **515**-*a*-**1** and a command channel of a second channel **505**-*a* that includes the pseudo-channel **515**-*a*-**3**. For example, command signaling may continue to be communicated via the command channel of the first channel **505**-*a* using the first interface block **220** in association with communicating data via another pseudo-channel **515**-*a* of the first channel **505**-*a*, and command signaling associated with the communication of data via the pseudo-channel **515**-*a*-**1** may instead be communicated via the command channel of the second channel **505**-*a* using the second interface block **220**. The logic block **230** may similarly configure a third interface block **220** associated with the pseudo-channel **515**-*a*-**2** and/or the second interface block **220** associated with the pseudo-channel **515**-*a*-**4** to communicate shared command signaling via a command channel of a channel **505**-*a* that includes the pseudo-channel **515**-*a*-**2** and a command channel of a channel **505**-*a* that includes the pseudo-channel **515**-*a*-**4**.
[0101] In some examples, a die **240** that includes a pseudo-channel **515** to which a portion of an address space is remapped (e.g., a pseudo-channel of a spare die **240**) may be located at various positions within a stack **425**. In some cases, a spare die **240** may be located at a bottom of the stack **425**, which may provide thermal benefits to other dies **240** of the stack **425**. For example, dies **240**-

c located nearer to the die **205**-*c*-**3** along the z-direction may be relatively hotter than dies **240**-*c* located further from the die **205**-*c*-**3** due to a high power density associated with the die **205**-*c*-**3**. By positioning a spare die **240**-*c* nearer to the die **205**-*c*-**3**, other dies **240**-*c* of the stack **425** may be relatively cooler, thereby improving performance of a nominal configuration, or decreasing the likelihood of errors that may result in the logic block **230** performing the remapping, or both.

[0102] FIG. **5**D shows an example of a sparing diagram **500**-*d* that supports bank-level sparing. For example, the sparing diagram **500**-*d* depicts a die **205**-*c*-**4** coupled with a set of dies **240**-*c* stacked along a direction (e.g., a z-direction), which may be an example of a stack **425**. An error associated with a die **240**-*c*-**8** of the set of dies **240**-*c* may be detected as described herein. For example, each die **240**-*c* may include one or more channels **505**-*a* via which one or more memory arrays **250** of the die **240**-*c* may be accessed. In some examples, a memory array **250** may include one or more banks **520**-*a*, which may be banks of memory cells. In some examples, a memory array **250**, or a portion thereof, may be an example of a bank **520**-*a*. In the example of FIG. **5**D, the die **240**-*c*-**8** may include a channel **505**-*a*-**3** via which one or more banks **520**-*a* may be accessed. For example, the channel **505**-*a*-**3** may be used to access banks **520**-*a*-**1**, **520**-*a*-**2**, **520**-*a*-**3**, and **520**-*a*-**4**. In the example of FIG. **5**D, an error associated with the bank **520**-*a*-**2** may be detected.

[0103] Based on the detection of the error, a logic block **230** of the die **205**-*c*-**4** may remap access of the bank **520**-*a*-**2** to another bank **520**-*a* of the set of dies **240**-*c*. For example, an address space available to a host system may include a portion of addresses that may be used by the host system to access the bank **520**-*a*-**2**. The logic block **230** may remap the portion of addresses such that use of addresses within the portion by the host system may instead access another bank **520**-*a*, which may be referred to as a spare bank **520**-*a*. For example, a die **240**-*c*-**9** of the set of dies **240**-*c* may include a channel **505**-*a*-**4** (e.g., included in a same channel set **510**-*a* that includes the channel **505**-*a*-**3**) that supports accessing banks **520**-*a*-**5**, **520**-*a*-**6**, **520**-*a*-**7**, and **520**-*a*-**8**. The logic block **230** may remap accessing from the bank **520**-*a*-**2** to the bank **520**-*a*-**6** or some other bank **520**-*a* of the channel **505**-*a*-**4** or the die **240**-*c*-**9**. Additionally, or alternatively, the die **240**-*c*-**8** may include one or more spare banks **520** (e.g., associated with one or more spare channels **505**-*a*), and the logic block **230** may remap access of the bank **520**-*a*-**2** to a spare bank **520**-*a* of the die **240**-*c*-**8**. In some examples, the die **240**-*c*-**9** may be a spare die of the set of dies **240**-*c*, and the banks **520**-*a* of the die **240**-*c*-**9** may be spare banks **520**-*a*.

[0104] To support remapping the portion of the address space, the logic block **230** may remap accessing of the bank **520**-*a*-**2** from a first interface block **220** of the die **205**-*c*-**4** associated with the channel **505**-*a*-**3** to a second interface block **220** of the die **205**-*c*-**4** associated with the channel **505**-*a*-**4** that includes the bank **520**-*a*-**6** or some other channel **505**-*a* that includes the bank **520**-*a* to which access of the bank **520**-*a*-**2** is remapped. For example, the portion of the address space corresponding to the bank **520**-*a*-**2** may be remapped to the second interface block **220** such that accesses of addresses within the remapped portion of the address space may be supported by the second interface block **220**. In some examples, the second interface block **220** may be referred to as a spare interface block **220** (e.g., based on being coupled with an interface block **245** associated with a spare channel **505**-*a*).

[0105] In some examples, a die **240** that includes a bank **520** to which a portion of an address space is remapped (e.g., a bank **520** of a spare die **240**) may be located at various positions within a stack **425**. In some cases, a spare die **240** may be located at a bottom of the stack **425**, which may provide thermal benefits to other dies **240** of the stack **425**. For example, dies **240**-*c* located nearer to the die **205**-*c*-**4** along the z-direction may be relatively hotter than dies **240**-*c* located further from the die **205**-*c*-**4** due to a high power density associated with the die **205**-*c*-**4**. By positioning a spare die **240**-*c* nearer to the die **205**-*c*-**4**, other dies **240**-*c* of the stack **425** may be relatively cooler, thereby improving performance of a nominal configuration, or decreasing the likelihood of errors that may result in the logic block **230** performing the remapping, or both.

[0106] FIG. **6** shows a block diagram **600** of a system **620** that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein. The system **620** may be an example of aspects of a system as described with reference to FIGS. **1** through **5**. The system **620**, or various components thereof (e.g., an interface block **220**, circuitry of an interface block **220**, an interface block **245**, circuitry of an interface block **245**), may be an example of means for performing various aspects of sparing techniques in stacked memory architectures as described herein. For example, the system **620** may include an evaluation component **625**, a non-volatile storage component **630**, a remap component **635**, a coupling component **640**, or any combination thereof. In some examples, at least a portion of the system **620** (e.g., at least a portion of the evaluation component **625**, at least a portion of the non-volatile storage component **630**, at least a portion of the remap component **635**, or a combination thereof) may be included in a semiconductor system, such as a system **200**. Additionally, or alternatively, at least a portion of the system **620** (e.g., at least a portion of the evaluation component **625**, at least a portion of the non-volatile storage component **630**, at least a portion of the remap component **635**, at least a portion of the coupling component **640**, or a combination thereof) may be implemented by a manufacturing system or one or more controllers associated with a manufacturing system. For example, in some cases, one or more operations performed by the evaluation component **625**, the non-volatile storage component **630**, the coupling component **640**, or a combination thereof, may be performed by a manufacturing system or one or more controllers associated with a manufacturing system. Each of these components, or components of subcomponents thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0107] The evaluation component **625** (e.g., of an interface block **220**, of an interface block **245**, of a logic block **230**, of a manufacturing system) may be configured as or otherwise support a means for evaluating function of each of a plurality of array dies (e.g., dies **240**). The non-volatile storage component **630** (e.g., of an interface block **220**, of an interface block **245**, of a logic block **230**, of a manufacturing system) may be configured as or otherwise support a means for setting, based on the evaluating, one or more non-volatile storage elements of the plurality of array dies (e.g., of a non-volatile storage **270**) to indicate an error associated with one or more of the plurality of array dies. The remap component **635** (e.g., of an interface block **220**, of a logic block **230**, of a manufacturing system) may be configured as or otherwise support a means for remapping access of the plurality of array dies, using logic circuitry (e.g., of a logic block **230**) of a logic die (e.g., a die **205**) that is coupled with the plurality of array dies, from one or more first memory arrays (e.g., memory arrays **250**, channels **505**, pseudo-channels **515**, banks **520**) of the plurality of array dies to one or more second memory arrays (e.g., memory arrays **250**, channels **505**, pseudo-channels **515**, banks **520**) of the plurality of array dies based on setting the one or more non-volatile storage elements.

[0108] In some examples, the coupling component **640** may be configured as or otherwise support a means for coupling, after evaluating the function of each of the plurality of array dies, the plurality of array dies in a stack along a direction (e.g., in a stack **410**).

[0109] In some examples, the coupling component **640** may be configured as or otherwise support a means for coupling the plurality of array dies in a stack along a direction (e.g., in a stack **410**), and evaluating the function of each of the plurality of array dies may be performed after the coupling.

[0110] In some examples, the coupling component **640** may be configured as or otherwise support a means for coupling the plurality of array dies and the logic die (e.g., in a stack **420**, in a stack **425**), and evaluating the function of each of the plurality of array dies may be performed after the coupling.

[0111] In some examples, to support setting the one or more non-volatile storage elements, the non-volatile storage component **630** may be configured as or otherwise support a means for setting one or more one-time programmable storage elements (e.g., fuses, antifuses, or a combination thereof) of the plurality of array dies to indicate the error.

[0112] In some examples, to support remapping access of the plurality of array dies, the remap component **635** may be configured as or otherwise support a means for remapping accessing from a first array die of the plurality of array dies that includes the one or more first memory arrays to a second array die of the plurality of array dies that includes the one or more second memory arrays.

[0113] In some examples, to support remapping access of the plurality of array dies, the remap component **635** may be configured as or otherwise support a means for remapping access from a first interface (e.g., a first interface block **245**) of an array die of the plurality of array dies, the first interface associated with accessing the one or more first memory arrays, to a second interface (e.g., a second interface block **245**) of the array die, the second interface associated with accessing the one or more second memory arrays.

[0114] In some examples, the logic die includes a plurality of second interfaces (e.g., interface blocks **220**) each associated with a respective plurality of channels (e.g., channels **505**, buses **221**, buses **246**, buses **255**, buses **301**, buses **302**, buses **303**, buses **304**) for communications with a respective first interface (e.g., an interface block **245**) of one of the plurality of array dies, the respective plurality of channels including a command channel (e.g., a bus **301**) and a plurality of data channels (e.g., buses **303**) operable to communicate data with at least one memory array corresponding to the respective first interface. In some examples, to support remapping access of the plurality of array dies, the remap component **635** may be configured as or otherwise support a means for remapping accessing from a first data channel of the respective plurality of channels associated with one second interface to a second data channel of the respective plurality of channels associated with another second interface.

[0115] In some examples, each array die of the plurality of array dies includes one or more memory arrays that each include one or more banks of memory cells (e.g., banks **520**). In some examples, to support remapping access of the plurality of array dies, the remap component **635** may be configured as or otherwise support a means for remapping access from one or more first banks included in the one or more first memory arrays to one or more second banks included in the one or more second memory arrays.

[0116] In some examples, the described functionality of the system **620**, or various components thereof, may be supported by or may refer to at least a portion of at least one processor, where such at least one processor may include one or more processing elements (e.g., a controller, a microprocessor, a microcontroller, a digital signal processor, a state machine, discrete gate logic, discrete transistor logic, discrete hardware components, or any combination of one or more of such elements). In some examples, the described functionality of the system **620**, or various components thereof, may be implemented at least in part by instructions (e.g., stored in memory, non-transitory computer-readable medium) executable by such at least one processor.

[0117] FIG. **7** shows a block diagram **700** of a logic die **720** that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein. The logic die **720** may be an example of aspects of a logic die (e.g., a die **205**) as described with reference to FIGS. **1** through **5**. The logic die **720**, or various components thereof (e.g., an interface block **220**, a logic block **230**), may be an example of means for performing various aspects of sparing techniques in stacked memory architectures as described herein. For example, the logic die **720** (e.g., an interface block **220**, a logic block **230**, or a combination thereof) may include an error component **725**, a remap component **730**, an access component **735**, or any combination thereof. Each of these components, or components of subcomponents thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0118] The error component **725** may be configured as or otherwise support a means for detecting, by logic circuitry (e.g., a logic block **230**) of a logic die (e.g., a die **205**) stacked with a plurality of array dies (e.g., dies **240**), an error associated with one or more first memory arrays (e.g., memory arrays **250**, channels **505**, pseudo-channels **515**, banks **520**) of the plurality of array dies. The remap component **730** may be configured as or otherwise support a means for remapping, by the logic circuitry based on detecting the error, access of the plurality of array dies from the one or more first memory arrays using a first interface (e.g., a first interface block **220**) of the logic die to one or more second memory arrays (e.g., memory arrays **250**, channels **505**, pseudo-channels **515**, banks **520**) of the plurality of array dies using a second interface (e.g., a second interface block **220**) of the logic die. The access component **735** may be configured as or otherwise support a means for accessing, using the second interface, the one or more second memory arrays based on the remapping.

[0119] In some examples, to support detecting the error, the error component **725** may be configured as or otherwise support a means for accessing non-volatile storage (e.g., non-volatile storage **235**, non-volatile storage **270**) associated with the one or more first memory arrays, where one or more non-volatile storage elements of the non-volatile storage indicate the error.

[0120] In some examples, to support detecting the error, the error component **725** may be configured as or otherwise support a means for determining the error associated with the one or more first memory arrays based on an attempt to access the one or more first memory arrays using the first interface of the logic die.

[0121] In some examples, to support remapping access of the plurality of array dies, the remap component **730** may be configured as or otherwise support a means for remapping accessing from a first array die of the plurality of array dies that includes the one or more first memory arrays to a second array die of the plurality of array dies that includes the one or more second memory arrays.

[0122] In some examples, to support remapping access of the plurality of array dies, the remap component **730** may be configured as or otherwise support a means for remapping access from a third interface (e.g., a first interface block **245**) of an array die of the plurality of array dies, the third interface associated with accessing the one or more first memory arrays, to a fourth interface (e.g., a second interface block **245**) of the array die, the fourth interface associated with accessing the one or more second memory arrays.

[0123] In some examples, the first interface and the second interface are each associated with a respective plurality of channels (e.g., channels **505**), the respective plurality of channels including a command channel (e.g., a bus **301**) and a plurality of data channels (e.g., buses **303**) operable to communicate data with at least one memory array corresponding to the interface. In some examples, remapping access of the plurality of array dies includes remapping access from a first data channel of the respective plurality of channels associated with the first interface to a second data channel of the respective plurality of channels associated with the second interface.

[0124] In some examples, each array die of the plurality of array dies includes one or more memory arrays that each include one or more banks of memory cells (e.g., banks **520**). In some examples, remapping access of the plurality of array dies includes remapping access from one or more first banks included in the one or more first memory arrays to one or more second banks included in the one or more second memory arrays.

[0125] In some examples, the described functionality of the logic die **720**, or various components thereof, may be supported by or may refer to at least a portion of at least one processor, where such at least one processor may include one or more processing elements (e.g., a controller, a microprocessor, a microcontroller, a digital signal processor, a state machine, discrete gate logic, discrete transistor logic, discrete hardware components, or any combination of one or more of such elements). In some examples, the described functionality of the logic die **720**, or various components thereof, may be implemented at least in part by instructions (e.g., stored in memory, non-transitory computer-readable medium) executable by such at least one processor.

[0126] FIG. **8** shows a flowchart illustrating a method **800** that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein. The operations of method **800** may be implemented by a system or its components as described herein (e.g., a die **205**, a die **240**, a logic block **230**). For example, the operations of method **800** may be performed by a system as described with reference to FIGS. **1** through **6**. In some examples, a system may execute a set of instructions to control the functional elements of the device to perform the described functions. Additionally, or alternatively, the system may perform aspects of the described functions using special-purpose hardware. In some examples, one or more operations of method **800** may be implemented by a manufacturing system or one or more controllers associated with a manufacturing system. In some examples, one or more controllers may execute a set of instructions to control one or more functional elements of the manufacturing system to perform the described functions. Additionally, or alternatively, one or more controllers may perform aspects of the described functions using special-purpose hardware.

[0127] At **805**, the method may include evaluating function of each of a plurality of array dies. In some examples, aspects of the operations of **805** may be performed by an evaluation component **625** as described with reference to FIG. **6**. For example, respective functions of dies **240** of a die stack (e.g., in a stack **410**, in a stack **420**, in a stack **425**) may be evaluated. In some examples, aspects of the operations of **805** may be performed by a manufacturing system or one or more controllers associated with a manufacturing system.

[0128] At **810**, the method may include setting, based on the evaluating, one or more non-volatile storage elements of the plurality of array dies to indicate an error associated with one or more of the plurality of array dies. In some examples, aspects of the operations of **810** may be performed by a non-volatile storage component **630** as described with reference to FIG. **6**. For example, non-volatile storage elements of one or more non-volatile storages **270** or a non-volatile storage **235** may be set (e.g., by an interface block **220**, by an interface block **245**, by a logic block **230**) to indicate an error associated with one or more of the dies **240** of the stack. In some examples, aspects of the operations of **810** may be performed by a manufacturing system or one or more controllers associated with a manufacturing system.

[0129] At **815**, the method may include remapping access of the plurality of array dies, using logic circuitry of a logic die that is coupled with the plurality of array dies, from one or more first memory arrays of the plurality of array dies to one or more second memory arrays of the plurality of array dies based on setting the one or more non-volatile storage elements. For example, a logic block **230** of a die **205** coupled with the dies **240** may remap accessing of the dies **240** from one or more first memory arrays (e.g., memory arrays **250**, channels **505**, pseudo-channels **515**, banks **520**) of the dies **240** to one or more second memory arrays (e.g., memory arrays **250**, channels **505**, pseudo-channels **515**, banks **520**) of the dies **240** based on setting the non-volatile storage elements of the one or more non-volatile storages **270**. In some examples, aspects of the operations of **815** may be performed by a remap component **635** as described with reference to FIG. **6**.

[0130] In some examples, an apparatus as described herein may perform a method or methods, such as the method **800**. The apparatus may include features, circuitry, logic, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor), or any combination thereof for performing the following aspects of the present disclosure. In some examples, one or more the following aspects may be performed by a manufacturing system that may include features, circuitry, logic, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by one or more controllers to control one or more functional elements of the manufacturing system), or any combination thereof for performing the one or more aspects.

[0131] Aspect 1: A method, apparatus (e.g., a manufacturing system), or non-transitory computer-readable medium including operations, features, circuitry, logic, means, or instructions, or any combination thereof for evaluating function of each of a plurality of array dies; setting, based on the evaluating, one or more non-volatile storage elements of the plurality of array dies to indicate an error associated with one or more of the plurality of array dies; and remapping access of the plurality of array dies, using logic circuitry of a logic die that is coupled with the plurality of array dies, from one or more first memory arrays of the plurality of array dies to one or more second memory arrays of the plurality of array dies based on setting the one or more non-volatile storage elements.

[0132] Aspect 2: The method, apparatus (e.g., manufacturing system), or non-transitory computer-readable medium of aspect 1, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for coupling, after evaluating the function of each of the plurality of array dies, the plurality of array dies in a stack along a direction.

[0133] Aspect 3: The method, apparatus (e.g., manufacturing system), or non-transitory computer-readable medium of any of aspects 1 through 2, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for coupling the plurality of array dies in a stack along a direction, where evaluating the function of each of the plurality of array dies is performed after the coupling.

[0134] Aspect 4: The method, apparatus (e.g., manufacturing system), or non-transitory computer-readable medium of any of aspects 1 through 3, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for coupling the plurality of array dies and the logic die, where evaluating the function of each of the plurality of array dies is performed after the coupling.

[0135] Aspect 5: The method, apparatus (e.g., manufacturing system), or non-transitory computer-readable medium of any of aspects 1 through 4, where setting the one or more non-volatile storage elements includes operations, features, circuitry, logic, means, or instructions, or any combination thereof for setting one or more one-time programmable storage elements of the plurality of array dies to indicate the error.

[0136] Aspect 6: The method, apparatus, or non-transitory computer-readable medium of any of aspects 1 through 5, where remapping access of the plurality of array dies includes operations, features, circuitry, logic, means, or instructions, or any combination thereof for remapping accessing from a first array die of the plurality of array dies that includes the one or more first memory arrays to a second array die of the plurality of array dies that includes the one or more second memory arrays.

[0137] Aspect 7: The method, apparatus, or non-transitory computer-readable medium of any of aspects 1 through 6, where remapping access of the plurality of array dies includes operations, features, circuitry, logic, means, or instructions, or any combination thereof for remapping access from a first interface of an array die of the plurality of array dies, the first interface associated with accessing the one or more first memory arrays, to a second interface of the array die, the second interface associated with accessing the one or more second memory arrays.

[0138] Aspect 8: The method, apparatus, or non-transitory computer-readable medium of any of aspects 1 through 7, where the logic die includes a plurality of second interfaces each associated with a respective plurality of channels for communications with a respective first interface of one of the plurality of array dies, the respective plurality of channels including a command channel and a plurality of data channels operable to communicate data with at least one memory array corresponding to the respective first interface, and remapping access of the plurality of array dies includes remapping accessing from a first data channel of the respective plurality of channels associated with one second interface to a second data channel of the respective plurality of channels associated with another second interface.

[0139] Aspect 9: The method, apparatus, or non-transitory computer-readable medium of any of aspects 1 through 8, where each array die of the plurality of array dies includes one or more memory arrays that each include one or more banks of memory cells, and remapping access of the plurality of array dies includes remapping access from one or more first banks included in the one or more first memory arrays to one or more second banks included in the one or more second memory arrays.

[0140] FIG. **9** shows a flowchart illustrating a method **900** that supports sparing techniques in stacked memory architectures in accordance with examples as disclosed herein. The operations of method **900** may be implemented by a logic die (e.g., a die **205**) or its components as described herein (e.g., an interface block **220**, a logic block **230**). For example, the operations of method **900** may be performed by a logic die as described with reference to FIGS. **1** through **5** and **7**. In some examples, a logic die may execute a set of instructions to control the functional elements of the device to perform the described functions. Additionally, or alternatively, the logic die may perform aspects of the described functions using special-purpose hardware.

[0141] At **905**, the method may include detecting, by logic circuitry of a logic die stacked with a plurality of array dies, an error associated with one or more first memory arrays of the plurality of array dies. For example, a logic block **230** of a die **205** stacked with a set of dies **240** (e.g., in a stack **425**) may detect an error associated with one or more first memory arrays (e.g., memory arrays **250**, channels **505**, pseudo-channels **515**, banks **520**) of the dies **240**. In some examples, aspects of the operations of **905** may be performed by an error component **725** as described with reference to FIG. **7**.

[0142] At **910**, the method may include remapping, by the logic circuitry based on detecting the error, access of the plurality of array dies from the one or more first memory arrays using a first interface of the logic die to one or more second memory arrays of the plurality of array dies using a second interface of the logic die. For example, based on detecting the error, the logic block **230** may remap access of the dies **240** from one or more first memory arrays (e.g., memory arrays **250**, channels **505**, pseudo-channels **515**, banks **520**) using a first interface block **220** of the die **205** to one or more second memory arrays (e.g., memory arrays **250**, channels **505**, pseudo-channels **515**, banks **520**) using a second interface block **220** of the die **205**. In some examples, aspects of the operations of **910** may be performed by a remap component **730** as described with reference to FIG. **7**.

[0143] At **915**, the method may include accessing, using the second interface, the one or more second memory arrays based on the remapping. For example, the second interface block **220** may access the one or more second memory arrays (e.g., memory arrays **250**, channels **505**, pseudo-channels **515**, banks **520**) based on the remapping. In some examples, aspects of the operations of **915** may be performed by an access component **735** as described with reference to FIG. **7**.

[0144] In some examples, an apparatus as described herein may perform a method or methods, such as the method **900**. The apparatus may include features, circuitry, logic, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor), or any combination thereof for performing the following aspects of the present disclosure:

[0145] Aspect 10: A method, apparatus, or non-transitory computer-readable medium including operations, features, circuitry, logic, means, or instructions, or any combination thereof for detecting, by logic circuitry of a logic die stacked with a plurality of array dies, an error associated with one or more first memory arrays of the plurality of array dies; remapping, by the logic circuitry based on detecting the error, access of the plurality of array dies from the one or more first memory arrays using a first interface of the logic die to one or more second memory arrays of the plurality of array dies using a second interface of the logic die; and accessing, using the second interface, the one or more second memory arrays based on the remapping.

[0146] Aspect 11: The method, apparatus, or non-transitory computer-readable medium of aspect 10, where detecting the error includes operations, features, circuitry, logic, means, or instructions, or any combination thereof for accessing non-volatile storage associated with the one or more first memory arrays, where one or more non-volatile storage elements of the non-volatile storage indicate the error.

[0147] Aspect 12: The method, apparatus, or non-transitory computer-readable medium of any of aspects 10 through 11, where detecting the error includes operations, features, circuitry, logic, means, or instructions, or any combination thereof for determining the error associated with the one or more first memory arrays based on an attempt to access the one or more first memory arrays using the first interface of the logic die.

[0148] Aspect 13: The method, apparatus, or non-transitory computer-readable medium of any of aspects 10 through 12, where remapping access of the plurality of array dies includes operations, features, circuitry, logic, means, or instructions, or any combination thereof for remapping accessing from a first array die of the plurality of array dies that includes the one or more first memory arrays to a second array die of the plurality of array dies that includes the one or more second memory arrays.

[0149] Aspect 14: The method, apparatus, or non-transitory computer-readable medium of any of aspects 10 through 13, where remapping access of the plurality of array dies includes operations, features, circuitry, logic, means, or instructions, or any combination thereof for remapping access from a third interface of an array die of the plurality of array dies, the third interface associated with accessing the one or more first memory arrays, to a fourth interface of the array die, the fourth interface associated with accessing the one or more second memory arrays.

[0150] Aspect 15: The method, apparatus, or non-transitory computer-readable medium of any of aspects 10 through 14, where the first interface and the second interface are each associated with a respective plurality of channels, the respective plurality of channels including a command channel and a plurality of data channels operable to communicate data with at least one memory array corresponding to the interface, and remapping access of the plurality of array dies includes remapping access from a first data channel of the respective plurality of channels associated with the first interface to a second data channel of the respective plurality of channels associated with the second interface.

[0151] Aspect 16: The method, apparatus, or non-transitory computer-readable medium of any of aspects 10 through 15, where each array die of the plurality of array dies includes one or more memory arrays that each include one or more banks of memory cells, and remapping access of the plurality of array dies includes remapping access from one or more first banks included in the one or more first memory arrays to one or more second banks included in the one or more second memory arrays.

[0152] It should be noted that the aspects described herein describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, portions from two or more of the methods may be combined.

[0153] A system is described. The following provides an overview of aspects of the apparatus as described herein:

[0154] Aspect 17: A system, including: a plurality of array dies stacked along a direction, each array die including: one or more memory arrays; and one or more first interfaces, each first interface including first circuitry operable to access at least one corresponding memory array of the one or more memory arrays; and a logic die coupled with the plurality of array dies, the logic die including: a plurality of second interfaces, each second interface including second circuitry operable to communicate access signaling with a respective first interface of the plurality of array dies to access the at least one memory array corresponding to the respective first interface; and logic circuitry operable to remap accessing of the plurality of array dies from one or more first memory arrays of the plurality of array dies using a first of the plurality of second interfaces to one or more second memory arrays of the plurality of array dies using a second of the plurality of second interfaces based on an error associated with access of the one or more first memory arrays.

[0155] Aspect 18: The system of aspect 17, where, to remap accessing of the plurality of array dies, the logic circuitry is operable to: remap accessing from one of the first interfaces of an array die of the plurality of array dies that includes the first circuitry operable to access the one or more first memory arrays to another of the first interfaces of the array die that includes the first circuitry operable to access the one or more second memory arrays.

[0156] Aspect 19: The system of any of aspects 17 through 18, where, to remap accessing of the plurality of array dies, the logic circuitry is operable to: remap accessing from a first array die of the plurality of array dies that includes the one or more first memory arrays to a second array die of the plurality of array dies that includes the one or more second memory arrays.

[0157] Aspect 20: The system of any of aspects 17 through 19, where, to remap accessing of the plurality of array dies, the logic circuitry is operable to: remap access from all of the first interfaces of a first array die of the plurality of array dies to respective first interfaces of a second array die of the plurality of array dies.

[0158] Aspect 21: The system of any of aspects 17 through 20, where: each second interface is associated with a respective plurality of channels for communications with the respective first interface, the respective plurality of channels including a command channel and a plurality of data channels operable to communicate data with the at least one memory array corresponding to the respective first interface; and, to remap accessing of the plurality of array dies, the logic circuitry is operable to remap accessing from a first data channel of the respective plurality of channels associated with one second interface to a second data channel of the respective plurality of channels associated with another second interface.

[0159] Aspect 22: The system of aspect 21, where the logic circuitry is further operable to: configure the one second interface, the other second interface, or both to communicate shared command signaling via the command channel associated with the one second interface and via the command channel associated with the other second interface.

[0160] Aspect 23: The system of any of aspects 17 through 22, where: each of the one or more memory arrays includes one or more banks of memory cells; and, to remap accessing of the plurality of array dies, the logic circuitry is operable to remap accessing from one or more first banks included in the one or more first memory arrays to one or more second banks included in the one or more second memory arrays.

[0161] Aspect 24: The system of any of aspects 17 through 23, where the logic circuitry is further operable to: identify the error associated with access of the one or more first memory arrays based on accessing a non-volatile storage of an array die that includes the one or more first memory arrays; and remap the accessing of the plurality of array dies from the one or more first memory arrays to the one or more second memory arrays based on the identification.

[0162] Aspect 25: The system of any of aspects 17 through 24, where the logic circuitry is further operable to: identify the error associated with access of the one or more first memory arrays based on accessing a non-volatile storage of an array die that includes the one or more second memory arrays; and remap the accessing of the plurality of array dies from the one or more first memory arrays to the one or more second memory arrays based on the identification.

[0163] Aspect 26: The system of any of aspects 17 through 25, where the logic circuitry is further operable to: determine the error associated with access of the one or more first memory arrays based on an attempt to access the one or more first memory arrays by one of the plurality of second interfaces; and remap the accessing of the plurality of array dies from the one or more first memory arrays to the one or more second memory arrays based on the determination.

[0164] Aspect 27: The system of aspect 26, where the logic circuitry is further operable to: store an indication of the determined error in a non-volatile storage of the logic die, or one or more of the array dies, or a combination thereof.

[0165] Aspect 28: The system of any of aspects 17 through 27, where, to remap the accessing of the plurality of array dies, the logic circuitry is operable to: remap a portion of an address space accessible by a host system from the one or more first memory arrays to the one or more second memory arrays.

[0166] Aspect 29: The system of any of aspects 17 through 28, where the one or more second memory arrays are included in an array die of the plurality of array dies that is located nearest to the logic die.

[0167] Aspect 30: The system of any of aspects 17 through 29, where the one or more second memory arrays are included in an array die of the plurality of array dies over which other array dies of the plurality of array dies are stacked.

[0168] A system is described. The following provides an overview of aspects of the apparatus as described herein:

[0169] Aspect 31: A system, including: a plurality of array dies stacked along a direction, each array die including one or more memory arrays and a non-volatile storage; and a logic die coupled with the plurality of array dies, the logic die including: a plurality of interfaces including circuitry operable to communicate access signaling to access memory arrays of the plurality of array dies; and logic circuitry coupled with the plurality of interfaces and operable to: identify an error associated with one or more first memory arrays of the plurality of array dies based on accessing the non-volatile storage of one or more of the plurality of array dies; and remap, based on identifying the error, accessing from the one or more first memory arrays using a first interface of the plurality of interfaces to one or more second memory arrays of the plurality of array dies using a second interface of the plurality of interfaces.

[0170] A system is described. The following provides an overview of aspects of the apparatus as described herein:

[0171] Aspect 32: A system, including: a plurality of array dies stacked along a direction, each array die including one or more memory arrays; and a logic die coupled with the plurality of array dies, the logic die including: a plurality of interfaces including circuitry operable to communicate access signaling to access memory arrays of the plurality of array dies; and logic circuitry coupled with the plurality of interfaces and operable to: detect an error based on an attempt to access one or more first memory arrays of the plurality of array dies using a first interface of the plurality of interfaces; and remap, based on detecting the error, accessing of the plurality of array dies from the one or more first memory arrays using the first interface to one or more second memory arrays of the plurality of array dies using a second interface of the plurality of interfaces.

[0172] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, or symbols of signaling that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof. Some drawings may illustrate signals as a single signal; however, the signal may represent a bus of signals, where the bus may have a variety of bit widths.

[0173] The terms “electronic communication,” “conductive contact,” “connected,” and “coupled” may refer to a relationship between components that supports the flow of signals between the components. Components are considered in electronic communication with (e.g., in conductive contact with, connected with, coupled with) one another if there is any electrical path (e.g., conductive path) between the components that can, at any time, support the flow of signals (e.g., charge, current, voltage) between the components. A conductive path between components that are in electronic communication with each other (e.g., in conductive contact with, connected with, coupled with) may be an open circuit or a closed circuit based on the operation of the device that includes the connected components. A conductive path between connected components may be a direct conductive path between the components or may be an indirect conductive path that includes intermediate components, such as switches, transistors, or other components. In some examples, the flow of signals between the connected components may be interrupted for a time, for example, using one or more intermediate components such as switches or transistors.

[0174] The term “isolated” may refer to a relationship between components in which signals are not presently capable of flowing between the components. Components are isolated from each other if there is an open circuit between them. For example, two components separated by a switch that is positioned between the components are isolated from each other when the switch is open. When a component isolates two components, the component may initiate a change that prevents signals from flowing between the other components using a conductive path that previously permitted signals to flow.

[0175] The term “coupling” (e.g., “electrically coupling”) may refer to condition of moving from an open-circuit relationship between components in which signals are not presently capable of being communicated between the components (e.g., over a conductive path) to a closed-circuit relationship between components in which signals are capable of being communicated between components (e.g., over the conductive path). When a component, such as a controller, couples other components together, the component may initiate a change that allows signals to flow between the other components over a conductive path that previously did not permit signals to flow.
[0176] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The detailed description includes specific details to provide an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form to avoid obscuring the concepts of the described examples.
[0177] In the appended figures, similar components or features may have the same reference label. Similar components may be distinguished by following the reference label by one or more dashes and additional labeling that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the additional reference labels.
[0178] The functions described herein may be implemented in hardware, software executed by a processing system (e.g., one or more processors, one or more controllers, control circuitry processing circuitry, logic circuitry), firmware, or any combination thereof. If implemented in software executed by a processing system, the functions may be stored on or transmitted over as one or more instructions (e.g., code) on a computer-readable medium. Due to the nature of software, functions described herein can be implemented using software executed by a processing system, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.
[0179] Illustrative blocks and modules described herein may be implemented or performed with one or more processors, such as a DSP, an ASIC, an FPGA, discrete gate logic, discrete transistor logic, discrete hardware components, other programmable logic device, or any combination thereof designed to perform the functions described herein. A processor may be an example of a microprocessor, a controller, a microcontroller, a state machine, or other types of processor. A processor may also be implemented as at least one of one or more computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).
[0180] As used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”
[0181] As used herein, including in the claims, the article “a” before a noun is open-ended and understood to refer to “at least one” of those nouns or “one or more” of those nouns. Thus, the terms “a,” “at least one,” “one or more,” “at least one of one or more” may be interchangeable. For example, if a claim recites “a component” that performs one or more functions, each of the individual functions may be performed by a single component or by any combination of multiple components. Thus, the term “a component” having characteristics or performing functions may refer to “at least one of one or more components” having a particular characteristic or performing a

particular function. Subsequent reference to a component introduced with the article "a" using the terms "the" or "said" may refer to any or all of the one or more components. For example, a component introduced with the article "a" may be understood to mean "one or more components," and referring to "the component" subsequently in the claims may be understood to be equivalent to referring to "at least one of the one or more components." Similarly, subsequent reference to a component introduced as "one or more components" using the terms "the" or "said" may refer to any or all of the one or more components. For example, referring to "the one or more components" subsequently in the claims may be understood to be equivalent to referring to "at least one of the one or more components."
[0182] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium, or combination of multiple media, that can be accessed by a computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, electrically erasable programmable read-only memory (EEPROM), optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium or combination of media that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a computer, or a processor.
[0183] The descriptions and drawings are provided to enable a person having ordinary skill in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to the person having ordinary skill in the art, and the techniques disclosed herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A system, comprising: a plurality of array dies stacked along a direction, each array die comprising: one or more memory arrays; and one or more first interfaces, each first interface comprising first circuitry operable to access at least one corresponding memory array of the one or more memory arrays; and a logic die coupled with the plurality of array dies, the logic die comprising: a plurality of second interfaces, each second interface comprising second circuitry operable to communicate access signaling with a respective first interface of the plurality of array dies to access the at least one memory array corresponding to the respective first interface; and logic circuitry operable to remap accessing of the plurality of array dies from one or more first memory arrays of the plurality of array dies using a first of the plurality of second interfaces to one or more second memory arrays of the plurality of array dies using a second of the plurality of second interfaces based on an error associated with access of the one or more first memory arrays.
2. The system of claim 1, wherein, to remap accessing of the plurality of array dies, the logic circuitry is operable to: remap accessing from one of the first interfaces of an array die of the plurality of array dies that includes the first circuitry operable to access the one or more first memory arrays to another of the first interfaces of the array die that includes the first circuitry operable to access the one or more second memory arrays.
3. The system of claim 1, wherein, to remap accessing of the plurality of array dies, the logic circuitry is operable to: remap accessing from a first array die of the plurality of array dies that includes the one or more first memory arrays to a second array die of the plurality of array dies that includes the one or more second memory arrays.
4. The system of claim 1, wherein, to remap accessing of the plurality of array dies, the logic circuitry is operable to: remap access from all of the first interfaces of a first array die of the plurality of array dies to respective first interfaces of a second array die of the plurality of array

dies.

5. The system of claim 1, wherein: each second interface is associated with a respective plurality of channels for communications with the respective first interface, the respective plurality of channels including a command channel and a plurality of data channels operable to communicate data with the at least one memory array corresponding to the respective first interface; and to remap accessing of the plurality of array dies, the logic circuitry is operable to remap accessing from a first data channel of the respective plurality of channels associated with one second interface to a second data channel of the respective plurality of channels associated with another second interface.

6. The system of claim 5, wherein the logic circuitry is further operable to: configure the one second interface, the other second interface, or both to communicate shared command signaling via the command channel associated with the one second interface and via the command channel associated with the other second interface.

7. The system of claim 1, wherein: each of the one or more memory arrays comprises one or more banks of memory cells; and to remap accessing of the plurality of array dies, the logic circuitry is operable to remap accessing from one or more first banks included in the one or more first memory arrays to one or more second banks included in the one or more second memory arrays.

8. The system of claim 1, wherein the logic circuitry is further operable to: identify the error associated with access of the one or more first memory arrays based on accessing a non-volatile storage of an array die that includes the one or more first memory arrays; and remap the accessing of the plurality of array dies from the one or more first memory arrays to the one or more second memory arrays based on the identification.

9. The system of claim 1, wherein the logic circuitry is further operable to: identify the error associated with access of the one or more first memory arrays based on accessing a non-volatile storage of an array die that includes the one or more second memory arrays; and remap the accessing of the plurality of array dies from the one or more first memory arrays to the one or more second memory arrays based on the identification.

10. The system of claim 1, wherein the logic circuitry is further operable to: determine the error associated with access of the one or more first memory arrays based on an attempt to access the one or more first memory arrays by one of the plurality of second interfaces; and remap the accessing of the plurality of array dies from the one or more first memory arrays to the one or more second memory arrays based on the determination.

11. The system of claim 10, wherein the logic circuitry is further operable to: store an indication of the determined error in a non-volatile storage of the logic die, or one or more of the array dies, or a combination thereof.

12. The system of claim 1, wherein, to remap the accessing of the plurality of array dies, the logic circuitry is operable to: remap a portion of an address space accessible by a host system from the one or more first memory arrays to the one or more second memory arrays.

13. The system of claim 1, wherein the one or more second memory arrays are included in an array die of the plurality of array dies that is located nearest to the logic die.

14. The system of claim 1, where the one or more second memory arrays are included in an array die of the plurality of array dies over which other array dies of the plurality of array dies are stacked.

15. A method, comprising: evaluating function of each of a plurality of array dies; setting, based on the evaluating, one or more non-volatile storage elements of the plurality of array dies to indicate an error associated with one or more of the plurality of array dies; and remapping access of the plurality of array dies, using logic circuitry of a logic die that is coupled with the plurality of array dies, from one or more first memory arrays of the plurality of array dies to one or more second memory arrays of the plurality of array dies based on setting the one or more non-volatile storage elements.

16. The method of claim 15, further comprising: coupling, after evaluating the function of each of the plurality of array dies, the plurality of array dies in a stack along a direction.
17. The method of claim 15, further comprising: coupling the plurality of array dies in a stack along a direction, wherein evaluating the function of each of the plurality of array dies is performed after the coupling.
18. The method of claim 15, further comprising: coupling the plurality of array dies and the logic die, wherein evaluating the function of each of the plurality of array dies is performed after the coupling.
19. The method of claim 15, wherein setting the one or more non-volatile storage elements comprises: setting one or more one-time programmable storage elements of the plurality of array dies to indicate the error.
20. The method of claim 15, wherein remapping access of the plurality of array dies comprises: remapping accessing from a first array die of the plurality of array dies that includes the one or more first memory arrays to a second array die of the plurality of array dies that includes the one or more second memory arrays.
21. The method of claim 15, wherein remapping access of the plurality of array dies comprises: remapping access from a first interface of an array die of the plurality of array dies, the first interface associated with accessing the one or more first memory arrays, to a second interface of the array die, the second interface associated with accessing the one or more second memory arrays.
22. The method of claim 15, wherein: the logic die comprises a plurality of second interfaces each associated with a respective plurality of channels for communications with a respective first interface of one of the plurality of array dies, the respective plurality of channels including a command channel and a plurality of data channels operable to communicate data with at least one memory array corresponding to the respective first interface; and remapping access of the plurality of array dies comprises remapping accessing from a first data channel of the respective plurality of channels associated with one second interface to a second data channel of the respective plurality of channels associated with another second interface.
23. The method of claim 15, wherein: each array die of the plurality of array dies comprises one or more memory arrays that each comprise one or more banks of memory cells; and remapping access of the plurality of array dies comprises remapping access from one or more first banks included in the one or more first memory arrays to one or more second banks included in the one or more second memory arrays.
24. A system, comprising: a plurality of array dies stacked along a direction, each array die comprising one or more memory arrays and a non-volatile storage; and a logic die coupled with the plurality of array dies, the logic die comprising: a plurality of interfaces comprising circuitry operable to communicate access signaling to access memory arrays of the plurality of array dies; and logic circuitry coupled with the plurality of interfaces and operable to: identify an error associated with one or more first memory arrays of the plurality of array dies based on accessing the non-volatile storage of one or more of the plurality of array dies; and remap, based on identifying the error, accessing from the one or more first memory arrays using a first interface of the plurality of interfaces to one or more second memory arrays of the plurality of array dies using a second interface of the plurality of interfaces.
25. A method, comprising: detecting, by logic circuitry of a logic die stacked with a plurality of array dies, an error associated with one or more first memory arrays of the plurality of array dies; remapping, by the logic circuitry based on detecting the error, access of the plurality of array dies from the one or more first memory arrays using a first interface of the logic die to one or more second memory arrays of the plurality of array dies using a second interface of the logic die; and accessing, using the second interface, the one or more second memory arrays based on the remapping.

26. The method of claim 25, wherein detecting the error comprises: accessing non-volatile storage associated with the one or more first memory arrays, wherein one or more non-volatile storage elements of the non-volatile storage indicate the error.
27. The method of claim 25, wherein detecting the error comprises: determining the error associated with the one or more first memory arrays based on an attempt to access the one or more first memory arrays using the first interface of the logic die.
28. The method of claim 25, wherein remapping access of the plurality of array dies comprises: remapping accessing from a first array die of the plurality of array dies that includes the one or more first memory arrays to a second array die of the plurality of array dies that includes the one or more second memory arrays.
29. The method of claim 25, wherein remapping access of the plurality of array dies comprises: remapping access from a third interface of an array die of the plurality of array dies, the third interface associated with accessing the one or more first memory arrays, to a fourth interface of the array die, the fourth interface associated with accessing the one or more second memory arrays.
30. The method of claim 25, wherein: the first interface and the second interface are each associated with a respective plurality of channels, the respective plurality of channels including a command channel and a plurality of data channels operable to communicate data with at least one memory array corresponding to the interface; and remapping access of the plurality of array dies comprises remapping access from a first data channel of the respective plurality of channels associated with the first interface to a second data channel of the respective plurality of channels associated with the second interface.
31. The method of claim 25, wherein: each array die of the plurality of array dies comprises one or more memory arrays that each comprise one or more banks of memory cells; and remapping access of the plurality of array dies comprises remapping access from one or more first banks included in the one or more first memory arrays to one or more second banks included in the one or more second memory arrays.
32. A system, comprising: a plurality of array dies stacked along a direction, each array die comprising one or more memory arrays; and a logic die coupled with the plurality of array dies, the logic die comprising: a plurality of interfaces comprising circuitry operable to communicate access signaling to access memory arrays of the plurality of array dies; and logic circuitry coupled with the plurality of interfaces and operable to: detect an error based on an attempt to access one or more first memory arrays of the plurality of array dies using a first interface of the plurality of interfaces; and remap, based on detecting the error, accessing of the plurality of array dies from the one or more first memory arrays using the first interface to one or more second memory arrays of the plurality of array dies using a second interface of the plurality of interfaces.